

1MCo5

Construction Phase Plan Stage 1 Planned Works

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C12	29/08/2019	Peter Day	Neil Hancox	Daniel Altier	Updated for additional scope of works at South Portal
C13	17/09/2019	Peter Day	Neil Hancox	Daniel Altier	Internal review
C14	3/01/2020	Peter Day	Neil Hancox	Daniel Altier	Update on Geobind trial works, Prohibited types of plant/equipment, Pressurised Systems (part 10), Update on Chalfont St Peter Vent Shaft related works and in particular to the Access Road & Compound by our sub-contractor Roadbridge.
C15	20/01/2020	Peter Day	Neil Hancox	Daniel Altier	Update following Code 2 comment sheet on C14

SECURITY CLASSIFICATION: OFFICIAL

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1 Purpose

This Construction Phase (Health and Safety) Plan (CPP) sets out how work is to be planned, managed, monitored and coordinated on site during the Construction Phase in Sector C1 of the HS2 programme Phase 1. This plan will be further developed as the scope of work on site changes and new hazards are introduced.

This Construction Phase Plan includes HS2's requirements defined in the Works Information documents WI1100 Health and Safety [1]; WI1150 Occupational Health [2]; and WI825 Construction Management, which detail the Client's Supply Chain Health and Safety Standard for contractors working on the HS2 programme.

Each contractor working for ALIGN is required to develop their own Health and Safety Management Plan that states how they intend to meet the requirements of this document and any other legal and/or contractual obligations. The Health and Safety Management Plans will be reviewed and accepted by ALIGN as Principal Contractor for sector C1 before contractors may commence work.

Contractors shall ensure that this plan forms part of any contract it enters into with any other Contractor relevant to their work or services. Contractors shall inform ALIGN if they become aware or are aware of any breach of this plan by them or any third party.

2 Scope

This Construction Phase Plan applies to all construction work undertaken on the HS2 project on behalf of ALIGN. Initially the document relates to the Early Works carried out and it will be developed as the project progresses to cover all planned works.

Definitions and abbreviations

For the purposes of this document the following definitions shall apply:

#	Abbreviations	Definitions
[1]	WI 100	Description of the Works
[2]	WI 215	Security and Protection of Site
[3]	WI 218	Incident Management
[4]	WI235	Traffic and Transport
[5]	WI 285	Sustainability and Environmental Minimum Requirements
[6]	WI 300	Contractors Design
[7]	WI 880	Logistics
[8]	WI1100	WI1100 Health and Safety

#	Abbreviations	Definitions
[9]	Wl1150	Wl1150 Occupational Health
[10]	Safe@Heart	The strapline representing HS2's health and safety ethos and campaign
[11]	Temporary Works	Temporary works is defined in BS5975: 2008 "Code of practice for temporary works procedures and the permissible stress design of falsework" as "(those) parts of the works that allow or enable construction of, protect, support or provide access to, the permanent works and which might or might not remain in place at the completion of the works"
[12]	CCNSG	Client Contractor National Safety Group Safety Passport
[13]	CDM 2015	Construction Design and Management Regulations, 2015
[14]	CDMA	Construction Design and Management Advisor
[15]	CISRS	Construction Industry Scaffolders Record Scheme
[16]	CITB	Construction Industry Training Board
[17]	COSHH	Control of Substances Hazardous to Health
[18]	CPCS	Construction Plant Competence Scheme
[19]	CPP	Construction Phase (Health and Safety) Plan
[20]	CSCS	Construction Skills Competence Scheme
[21]	HAWS	Hand Arm Vibration Syndrome
[22]	HGV	Heavy Goods Vehicle
[23]	IPAF	International Powered Access Federation
[24]	MEWP	Mobile Elevating Work Platform
[25]	MSDS	Material Safety Data Sheet
[26]	NASC	National Access & Scaffolding Confederation
[27]	NEC	New Engineering Contract
[28]	NPORS	National plant Operators Registration Scheme
[29]	OH&S	Occupational Health and Safety
[30]	PASMA	Prefabricated Access Systems Manufacturers Association
[31]	PDF	Portable Document Format
[32]	PPE	Personal Protective Equipment
[33]	POWRA	Point of Work Risk Assessment
[34]	RAMS	Risk Assessment and Method Statement
[35]	RIDDOR	Reporting of Injuries, Diseases and Dangerous Occurrences Regulations 2013
[36]	SMS	Safety Management System
[37]	SSSTS	CITB Site Supervisor Safety Training Scheme (2-day course)
[38]	SMSTS	CITB Site Managers Safety Training Scheme (5-day course)

3 Description of Project

3.1 Location

The ALIGN JV consisting of Bouygues TP, Volker Fitzpatrick and Sir Robert McAlpine have been awarded the design and construction of HS2 sector C1, from Harvil Road in the Colne Valley near Harefield, Uxbridge, to the north portal of the Chiltern Tunnels near Great Missenden. The project involves the design and construction of the Colne Valley Viaduct, cut and fill earthworks, a culvert and side road overbridge, twin-bored tunnels through the Chiltern Hills with five ventilation/intervention shafts and cross passages every 350m. See Appendix 5 for a map of the project's location.

3.2 Works Description

Central Sector 1 (C1): Chiltern tunnels and Colne Valley viaduct - (25km 800m to 47km 750m). The headline assets for C1 run from 25km 800m at Colne Valley south embankment to 47km 750m north of Chiltern Tunnel North Portal, a distance of 21.950 kilometres. This Sector includes the Colne Valley viaduct, the longest viaduct in HS2 and when completed will be the longest built viaduct in the UK and the twin bored Chiltern tunnel which is 15.8 km in length.

ALIGN's sector of the project contains 18 Primary Assets, each of which is a significant element of infrastructure in its own right. ALIGN has contracted ALIGN-D to carry out the design of these Primary Assets and these are split into a number of packages. The following table is taken from the ALIGN-D Design Management Plan.

Asset No.	Primary Asset Description	Asset No.	Primary Asset Description
001009	Colne Valley South Embankment	000150	Chiltern Tunnel South Portal
001010	Colne Valley Viaduct	001014	Chiltern Tunnel South Cutting
000017	Ickenham ATFS	000151	Chiltern Tunnel
001011	Colne Valley North Embankment	000153	Chalfont St Peter Vent Shaft
001012	Tilehouse Lane Cutting	000154	Chalfont St Giles Vent Shaft and ATS
001013	West Hyde Embankment	000155	Amersham Vent Shaft
001278	Tilehouse Lane Overbridge	000156	Little Missenden Vent Shaft and ATS
001393	Shire Lane Culvert	020349	Chesham Road Vent Shaft
000180	West Hyde ATS	000157	Chiltern Tunnel North Portal

The project has been notified to the HSE as a construction project with Notification No. 54647A2D19. The notification will be updated via the Client as works progress with the latest revision being made in August 2019.

3.3 Early Works

The purpose of undertaking Early Works is to provide supplementary information for the safe and cost effective design and construction of the works contained within the Scope of Sector C1.

Additional scope of works involving bulk excavation works both temporary and permanent, formation of stockpiles and earth bunds has been appointed to ALIGN under the current early works contract.

HS2 Ltd have already completed a series of GI Projects, through a Framework of Ground Investigation Suppliers. Following on from this work, ALIGN has identified areas where supplementary information is required and therefore the first general intrusive construction works to be carried out by ALIGN will be ground investigation works. Works scheduled by ALIGN, as part of these early works and South Portal works, will include:

- Excavation and trial pits
- Testing to existing utility infrastructures.
- Bore Holes including monitoring
- Geobind trial works at Denham Water Ski Centre
- Utility Surveys - both intrusive and non-intrusive surveys
- Utility diversion and installation of temporary site services
- Topographical Surveys
- Land Contamination Surveys
- Ground stabilisation
- Ground Investigation Surveys (CPT, CP and Rotary)
- Bathymetry Surveys
- Ecological Surveys
- Noise and Vibration Surveys
- Compaction Tests
- Condition Surveys
- Asbestos Surveys
- Archaeological Surveys
- Removal of vegetation and trees
- Installation and testing of test piles
- Pumping test
- Noise monitoring
- Forming of site access to various locations
- Establishment of a site compound at Chalfont St Peter with access road and preparation to vent shaft activity.
- Establishment of works compound at Chalfont Lane to support main compound construction
- Vegetation/Tree clearance
- Traffic Management Measures
- Bulk earth work excavation at South Portal including both temporary and permanent works to include but not restricted to:

- Formation of tunnel cutting and work platforms, as to allow soil nailing works to be completed.
- Installation of soil nails to tunnel cutting and opening.
- Formation of stockpiles and bunds to South Portal as part of bulk excavation works including Chalfont Lane bund.
- Formation of Grout Block by the grouting of a grout platform at South Portal
- Formation of foundations for future Main Compound set up including production facilities, welfare facilities and associated civils works.

In addition to Ground Investigation works, works scheduled for 2019 include tree and shrub clearance in advance of the formation of piling platforms and compound establishments at 2 locations as follows:

- A412 Affinity Water Site (Load Test Pile 1)
- Harvil Road (Load Test Pile 2)

The above locations will be established as to allow Load Test Pile works to be undertaken, including Jetty Piles, CFA Piles, Grout Curtains, Sheet Piles and Main Load test piles. The test pile works will involve the construction activities of two piling contractors one carrying out Load test piles and Grout curtain installation. Other contractor's activity include jetty test pile installation and test on the installed pile. These two contractor will carry out work activities in one site at a time with continuous coordination from ALIGN JV.

The works at Load Test Pile 2 will include the removal and diversion works associated with the current 11KV running through the site. This, in summary, will involve the installation of new duct work, the re-routing of the overhead cables within the new ducts and removal of above ground poles. The 11KVA works will be undertaken by SSE appointed by ALIGN JV.

In addition at LTP 2 site, we will undertake works associated with a new temporary water supply, which will come from Harvil Road to the main site compound, and drainage works discharging into Newyears Green Bourne water course. Such works will be undertaken by Affinity Water for the water connection, with temporary water supply and drainage works being undertaken by ALIGN appointed contractor. These works will involve interface with other contractors, with such interfaces managed by ALIGN in accordance with the requirements of CDM and HS2 protocol, with all necessary consents and permissions being secured in advance of the scheduled works.

The new scope of works at the South Portal are due to commence early September 2019, with ALIGN appointed as Principal Contractor as defined under CDM. Initial works will be managed from the extended welfare provisions previously utilised for soil nail testing works. Upon the completion and handover of Compound Zero (early October 2019) being delivered by FUSION, the remaining scheduled works will be managed out of Compound Zero. In the interim, prior to occupation of compound Zero, there maybe be a requirement to establish a temporary works compound by KVJV to accommodate the Grout Platform installation works. Due to dual Principal Contractors working within the South Portal Site area, close co-ordination will be maintained through appointed Principal Contractors and Contractors, to ensure no additional risks are posed by scheduled works with particular focus upon deliveries and maintaining

unobstructed access/egress to emergency services in the event of an incident requiring the attendance of external emergency services does occur. It is currently envisaged that ALIGN will take sole Principal Contractors control of South Portal, upon completion of the remaining elements of FUSION current scope of works, including but not limited to, vegetation clearance, localised removal of surface ACM, installation of TBT electrical supply and maintenance of current access and main haul roads.

The required Early Works activities will take place according to the Early Works Programme. This programme will be updated and issued, the CPP will be revised if required to reflect additional scope.

3.4 Mobilisation

The main compound for the Early Works will be established at the location of the proposed vent shaft in Chalfont St Peter. This main compound will be further supported by satellite compounds at LTP1 and LTP 2. For the scheduled works at South Portal, the extended Soil Nail testing compound will be used until the occupation of Compound Zero (early October 2019). For the works, ALIGN is the Principal Contractor and is responsible for health and safety and for ensuring the appointed contractors are accountable for their own health and safety during the planned works. Risk Assessments and Method Statements will be provided in advance of any works commencing and be included in the RAMS register. No works will be allowed to commence until the submitted RAMS have been approved to Status A (Refer the ALIGN Method Statement Procedure - 1MCo5-ALJ-HS-PRO-Coo1-000008)

The scheduling of ALIGN work across the C1 sector area necessitates close working between ALIGN and the other contractors including, Fusion, EK, CSJV, SCS, Osborne, Cadent Gas, Affinity Water, National Grid, SEN, Highway England and Connect Plus. All organisations have respective responsibilities as Principal Contractor in their own areas of work and significant coordination and cooperation will be necessary to manage the interface. This will be managed through daily dialogue and weekly coordination meetings between the respective site teams.

3.5 Key Dates

Specific start and works duration details for the construction project are contained within the Early Works Programme (refer section 3.3)

4 Occupational Health and Safety Goals & Objectives

ALIGN and HS2 have common aspirations and goals in terms of health and safety. We all want to work together to build the high speed railway in a way that does not harm anyone now or in the future. HS2, as the Client and Principal Designer expresses its health and safety approach and values as shown below.



What does safety mean to us?



Safety is...
Caring for our workforce, our passengers and the public, by creating an environment where no one gets hurt.

Good looks like:



4.1 Key Performance Indicators

ALIGN is committed to a number of objectives under three headings of Organisation, Competence and Increasing the Sense of Responsibility of those involved. The detailed application of these objectives to ALIGN's work will be run alongside the HS2 Health and Safety Performance Indicators (HSPIs). HS2 will communicate their requirements.

An explanation of ALIGN's H&S objectives can be found in Appendix 1 as detailed within the ALIGN Health and Safety Policy Statement.

All contractors working in sector C1 are encouraged to introduce local reward and recognition schemes to promote and reflect our ethos of We Love Life and reinforce good safety behaviours, and to:

- Establish process objectives that are client focused and aligned to the objective of ALIGN.
- Establish Key Performance Indicators (KPIs) and the appropriate process of evaluation to determine if the process objectives are achieved.

4.2 ALIGN worker behaviour and engagement

ALIGN requires a high standard of safe working from its employees and requires the same high standards from its contractors. ALIGN will not tolerate poor safety behaviours within the office or at any of its site locations, and in order to develop the behaviours we want on the project, we will be rolling out our Behavioural Change programme. This programme is based on the Volker Fitzpatrick's Behavioural Change Programme, known as PALS.

4.2.1 The behavioural change programme seeks to be continually proactive in preventing all incidents and accidents throughout the duration of the project. The foundations of the programme are based around four basic rules:-

P – Plan – Plan the work, however insignificant it may seem. We do not start any work until everyone understands and agrees the Plan.

A – Attitude – Safety starts with the person and their Attitude. Every person needs to have the right Attitude to want to keep everyone safe.

L – Leadership – We all lead by example. Everyone with supervisory responsibility is a Leader and takes ownership of safety within their area of influence.

S – Share – we Share what we know and what we learn for the benefit of others and for the project as a whole.

All who work on the project are reminded of the programme's principles and objectives by attending briefings which are delivered on a quarterly basis. These briefings are bespoke and will be delivered by trained members of staff.

The subject matter of each briefing will vary from quarter to quarter but will always embrace matters and concerns surrounding Safety, Environment, Quality and Health.

This unique formula of constantly reminding everyone (little but often) of the importance of being vigilant at all times, empowers the workforce to speak up and challenge on all matters of concern. Continually communicating with the work force allows the project to both create and sustain a culture of value and worth where each individual is a committed member of the project team and not just a participant.

The behavioural change programme will be embraced and supported by all who work within the ALIGN joint venture allowing everyone to experience the excellence from being part of a great team that delivers and delivers successfully.

4.2.2 Health & Safety Basics

ALIGN will assist its contractors in any reasonably practicable way to facilitate safe working and to help achieve this, ALIGN has established 12 Health & Safety Basics or Fundamentaux. The 12 cover the following subjects:

1. Alcohol & Drugs
2. Personal Protective Equipment
3. Traffic
4. Pre-Task Briefing
5. Risk Analysis
6. Ergonomics
7. Stability
8. Hazardous Energy Control
9. Tools
10. Collective protection
11. Work at Height
12. Intervention



ALIGN intends that its sector of the HS2 project will be open, inclusive and transparent, with a positive "Just Safety Culture".

ALIGN's health and safety expectations and requirements placed upon Contractors are clearly communicated in this Construction Phase Plan and the HS2 "Supply Chain Health and Safety Standard." Both contractor organisations and individuals may be held accountable if they fail to comply with these requirements.

All ALIGN personnel and other contractors'/representatives must be made aware of the HS2 Supply Chain Health and Safety Standard. This document includes aspects to which personnel associated with the HS2 project should conduct themselves at work and within the local community.

4.2.3 Fair Culture

ALIGN's fair culture is called Aim High and is developed to include management of behaviour on site to ensure it meets ALIGN's expectations with regard to:

- a. Respect for good site practice and avoidance of horseplay
- b. Negligent acts
- c. Breach of Working Area rules
- d. Abusive behaviour.

Our positive safety culture is supported by our Aim High framework that secures the participation and engagement of our workforce, ensures expectations and accountabilities are clearly defined, and that all our stakeholders are treated fairly and equitably. This framework provides a charter to equitably manage safety behaviours, inclusive of the reward and recognition of positive safety behaviours, and the consistent and transparent management of any substantiated negative safety behaviours.

Our Just Culture Framework is defined by the following 5 pillars:

- **Informed Culture** - Persons are provided the necessary competency, information, instruction and training
- **Reporting Culture** - An organisational culture which fosters all persons proactively reporting all safety observation, dangerous occurrence, incident or hazard and best practices.
- **Just Culture** - Trust within our organisation that safety matters will be actioned; underpinned by the two way transfer of safety critical information
- **Flexible Culture** - People have the necessary competency and level of risk awareness to be flexible, adaptable and vigilant to manage change
- **Learning Culture** - Organisational transparency required to learn from safety incidents or failure

4.2.4 Communicating health and safety concerns

All construction personnel are permitted to stop work if they feel at risk. All staff have a responsibility to raise concerns about Health & Safety issues and where necessary stop work immediately if they consider they are faced with a situation of serious or imminent danger. All construction personnel are expected to raise concerns about Health & Safety issues to their supervisors or at the Daily Briefings and/or Toolbox Talks.

4.2.5 Safety Observation Reports

ALIGN operates a safety observation report system whereby all staff and contractors are able and encouraged to report any safety related observations that may have occur Usually the observations relate to unsafe situations or near misses, which are reported by means of a Safety Observation card that is completed and placed in a Safety Observation Box at work locations, or a suggestion box near reception at Euston House/Uxbridge Charter Building. All such submissions are to be reviewed by the site team and escalated appropriately for resolution.

Suspension of Works

Representatives of ALIGN can stop the works at any time in the event of a breach of health or safety requirements in order to minimise the potential for injury or ill-health to members of the workforce. All employees will be notified of the Suspension of Works Notification procedure at the site induction. ALIGN will notify HS2 when the Suspension of Works Notification procedure is invoked.

5 CDM Duty Holders

5.1 Client

Organisation	High Speed Two Limited
Representative	Paul Brown
Address	The Podium – 1 Eversholt Street, London, NW1 2DN
Tel No	0792 0594 601
Email	Paul.Brown@hs2.org.uk

5.2 Principal Designer

Organisation	High Speed Two Limited
Representative	Alasdair Hassan
Address	The Podium – 1 Eversholt Street, London, NW1 2DN
Tel No	07825 420129
Email	Alasdair.hassan@hs2.org.uk

5.3 Principal Contractor

Organisation	ALIGN JV
Representative	Daniel Altier
Address	Euston House 24 Eversholt Street London NW1 1AD
Tel No	07917 271149
Email	d.altier@alignjv.com

5.4 Designers

Organisation	ALIGN JV
Representative	Daniel Altier
Address	Euston House 24 Eversholt Street London NW1 1AD
Tel No	07917 271149
email	d.altier@alignjv.com

Organisation	ALIGN-D
Representative	Alan Price
Address	Euston House 24 Eversholt Street London NW1 1AD
Tel No	07718 862612
email	alan.price@jacobs.com

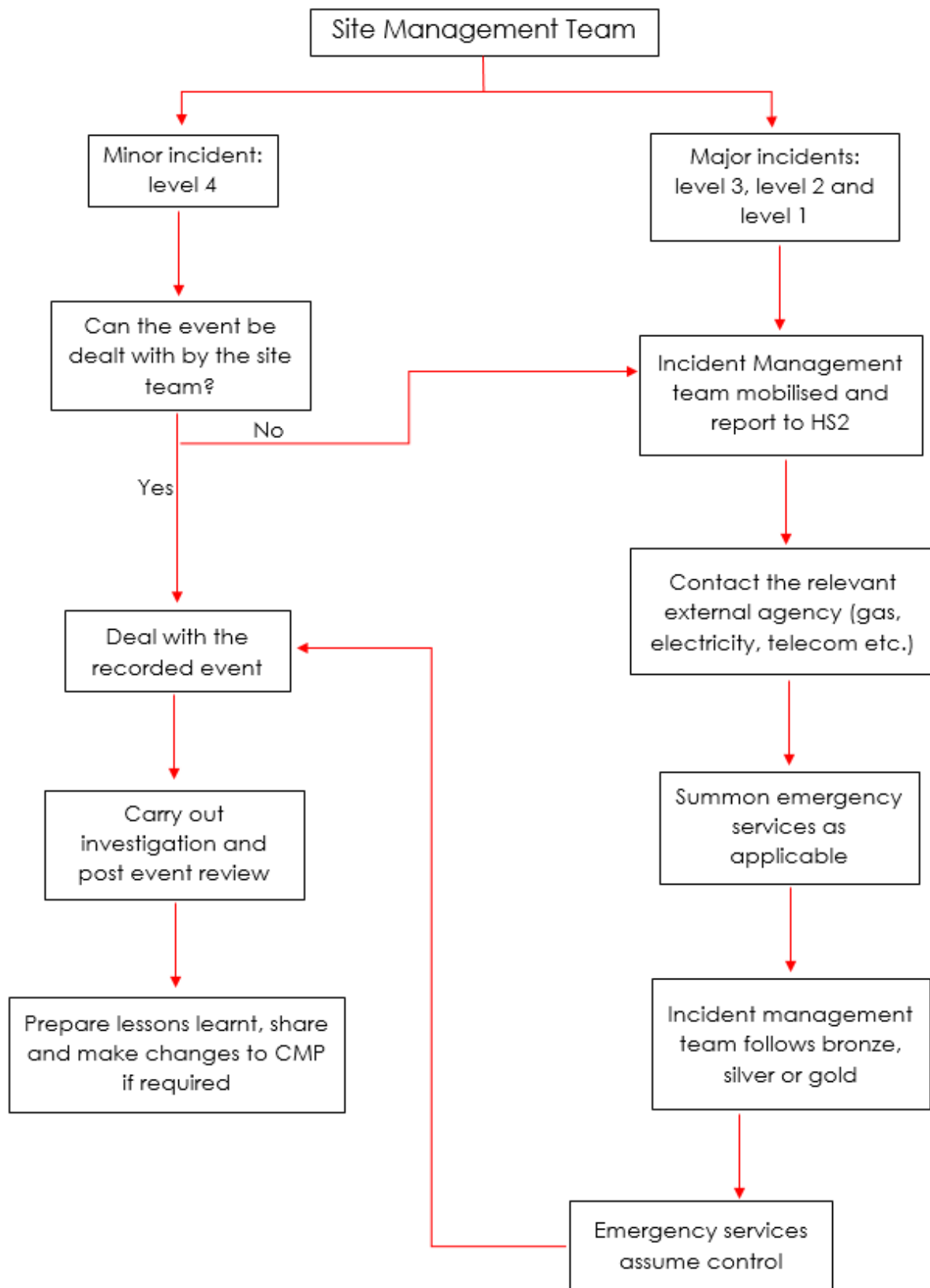
6 Directory

6.1 HS2 Complaints Number - 0808 1434 434

6.1.1 Emergencies

The HS2 Incident Management Procedure (HS2-HS2-CL-PLN-000-000001) [11] contains a definition of the terms used.

ALIGN's Site Emergency Preparedness and Response Plan (1MCo5-ALJ-HS-PLN-C001-000032), provides the details applicable to our Early Works packages. The process regarding response and reporting of incidents is summarised in the following flow chart.



6.2 ALIGN Project Office

Position	Project Director
Contact Name	Daniel Altier
Address	Euston House 24 Eversholt Street London NW1 1AD
Tel No	07917 271149
email	d.altier@alignjv.com

6.2.1 HS2 Health & Safety

HS2's health and safety team for sector C1 is co-located with ALIGN in Euston House.

The contact is Sarah Prince (Sarah.prince@hs2.org.uk)

HS2's health team is led by Fiona King who is based at the Snow Hill office in Birmingham (Fiona.king@hs2.org.uk)

6.2.2 Health & Safety Executive Construction Division

Rose Court

2 Southwark Bridge

London

SE1 9HS

[Rose Court street map](#) 

6.2.3 Bedford Division (covers Bucks area)

Woodlands

Manton Lane

Manton Industrial Estate

Bedford MK41 7LW

[Bedford street map](#) 

6.2.4 London Borough of Hillingdon

Civic Centre

High Street

Uxbridge

Middlesex

UB8 1UW

01895 250111

contact@hillingdon.gov.uk

6.2.5 Three Rivers District Council

Northway

Rickmansworth

WD3 1RL

01923 776611

www.threerivers.gov.uk

- 6.2.6 Chiltern District Council
King George V House
King George V Road
Amersham
HP6 5AW
01494 729000
www.chiltern.gov.uk
- 6.2.7 South Bucks District Council
Capswood
Oxford Road
Denham
UB9 4LH
Tel: 01895 837200
Mondays - Thursdays 8.45am - 5.15pm, Fridays - 8.45 am to 4.45pm
E-mail: JointCSTeam@southbucks.gov.uk
- 6.2.8 Police
Hillingdon Police Station
122-123 High Street
Hillingdon
UB8 1PG
- 6.2.9 Ambulance Service
London Ambulance Service
Royal Lane
Hillingdon
UB8 3QX
- 6.2.10 Fire & Rescue Service
London Fire Brigade
Uxbridge Road
Hillingdon
UB10 0PH
- 6.2.11 Hospitals
The nearest hospitals to the route with A&E Facilities are:
- Hillingdon Hospital**
Pield Heath Rd
Uxbridge
Middlesex
UB8 3NN
www.thh.nhs.uk
01895 238282

Stoke Mandeville Hospital

Mandeville Road

Aylesbury

Buckinghamshire

HP21 8AL

<http://www.buckshealthcare.nhs.uk/For%2opatients%2oand%2ovisitors/stoke-mandevillehospital>

01296 315000

7 Pre-construction Information

Pre-construction Information prepared by the Principal Designer, on behalf of the Client, will be formally transmitted to the relevant contractors by the ALIGN procurement team as part of invitations to tender and included in contracts. Confirmation that the contractor has received the most current pre-construction information will be made by the ALIGN Project Manager (or Construction Lead as appropriate) at on-site kick-off meetings and construction readiness reviews.

Pre-construction information for the project, known as the PCIP (Pre-construction Information Pack), has been prepared by the Principal Designer (HS2) and the current version is available through Business Collaborator (P1C-HS2-HS-REP-C001-000001).

7.1.1 Identification of hazards and risks

Key Safety Hazards and Risks have been identified in the Pre-construction Information Pack. This information will be supplemented by the Health & Safety File information provided by the previous GI campaigns carried out directly for HS2.

ALIGN-D, the design Joint Venture working on behalf of ALIGN, has developed the HS2 Design Risk Registers for each package of work within sector C1. The Design Risk Register for each relevant package, including the route-wide aspects, must be consulted when drawing up the specific Risk Assessments and Method Statements. In addition, the Project H&S Risk Register, through HORACE, in the form of HS2 Hazard Profile template.

The Design Risk Registers and Project H&S Risk Register are live documents that are regularly reviewed and updated following regular design review and H&S meetings. It is therefore important that the latest version of each document is consulted, and previously saved versions are not relied upon.

Historic information regarding utilities, buried and overhead services is included in the PCIP but must not be considered up to date. Current information on these utility services must be obtained through ALIGN's Utilities Manager, prior to the issue of any Permit to Dig system, Permit to work below overhead utilities and Permit to Cross Buried utilities.

Security threats are constantly evolving and therefore ALIGN's Security Manager must always be consulted when drawing up plans for compounds, work sites and making travel arrangements.

Construction compounds must follow the requirements laid out in CoCP.

The adjacent land usage throughout the Sector C1 is predominantly agricultural with occasional habitation and the occasional building(s) alongside.

The sector crosses and runs alongside a number of highway routes, watercourses, railways lines, third party assets and Areas of Outstanding Natural Beauty (AONB) and archaeological interest. Care must be taken at these locations to ensure compliance with regulations and procedures.

The HS2 project is very environmentally sensitive and there are numerous requirements laid down by the client and associated regulators to protect the environment during construction and use of the railway. These are laid out in the Works Information and managed by ALIGN's environmental team, Contractors must refer to the Projects Environmental Management Plan when planning works under their direction.

8 Management Arrangements

8.1 General Responsibilities

ALIGN's overall responsibility for Occupational Health and Safety management as Principal Contractor in Sector C1 rests with the Project Director, supported by the Director Responsible for Health and Safety and Occupational Health. The Principal Contractor's representative on site is named in the table in section 9.2.

Each individual working on HS2 has a degree of responsibility relative to their position within the organisation. The Project Director will ensure that each person working on the project, including contractors' personnel, are made aware of their individual responsibilities detailed under the ALIGN Health and Safety Policy and the HS2 ethos of Safe@heart.



"Safety is at the heart of everything we do"

"We will build safety into the heart of this railway"

"We will work at the heart of communities to protect their safety and wellbeing"

"Our people are at the heart of everything we do – we care about your health and wellbeing"

Safety leadership will be visibly demonstrated on the HS2 site. In particular, ALIGN will ensure that all front-line supervision and management are:

- Clear about their accountability and individual responsibility for OH&S

- Clear about the part human behaviour have in accident causation and what needs to be done to overcome this
- Clear that they have the right to stop work when they believe safety is compromised
- Clear that they never walk past an unsafe act or unsafe condition and that to do so is an unsafe act in itself
- Given skills in observation, intervention and feedback
- Clear about their own leadership style and how that can influence the performance of a team
- Clear what are the acceptable and unacceptable behaviours
- Involved and engaged in the safety communication process and take an active role in Daily Briefings, Pre-job briefings, Tool Box Talks, Feedback Sessions, etc.; and
- Engaged in the reward and recognition that celebrates our successes.

All line management on site – managers (SMSTS or NEBOSH Construction), supervisors /non-working foremen (SSSTS/IOSH Managing Safety minimum) and charge-hands/gangers (working foremen SSSTS /IOSH Managing Safety minimum qualification) – are key to ensuring that we achieve our Health & Safety Objectives through our ethos of We Love Life on this project. Line management must be experienced and competent within their specific discipline and must demonstrate positive leadership qualities by:

- Demonstrating a personal commitment to health and safety matters by means of their attitude and behaviours.
- Challenging unsafe behaviours or conditions and personally taking action to ensure that they are corrected.
- Ensuring that anyone working for them is aware of the agreed safe system of work.
- Ensuring that they fully understand the emergency arrangements for the site and are prepared to take a proactive role in the case of an emergency.
- Supporting any health and safety initiatives that are implemented across the project, by both personal participation and promotion within their respective organisation.

ALIGN expects the personal commitment of our employees and those from all designers, contractors, suppliers and consultants working in sector C1 in applying ALIGN's We Love Life principles of occupational health and safety:

- Every job will be done safely, no matter how important or urgent it is.
- Each of us has a personal responsibility for our own health and safety and for those around us.
- Putting people to work carries a specific responsibility and accountability for safety and health which will be visibly demonstrated.
- Each 'near miss' we learn from reduces the chance of harm next time.
- Each of us will spot, report and deal with hazards to help create a harm free workplace.

ALIGN promotes five mandatory simple actions that will allow all of us to focus on the way we can put our health and safety principles into practice on a day to day basis. Our simple actions are:

- I always address unsafe behaviours

- I always attend work fit for duty
- I always look where I am going to prevent slips, trips and falls
- I always hold the handrail on stairs
- I always keep my workplace tidy and free from obstructions

8.2 Key Principal Contractor Roles

The key Principal Contractor health and safety-related appointments have been made by the Project Director as listed below for the C1 Early Works. Additional roles will be added as the project scope develops.

Duty Role	Duty Post Holder
Project Director	Daniel Altier
Project Manager - Early Works	Clement Chasset
Project Manager - South Portal/ Shafts	Steve Meecham
Project Manager - South Portal	Mike Ohare
Health and Safety Director	Neil Hancox
Health and Safety Manager	Peter Day
Technical and Innovation Director	Phillippe Casteleyn
Logistics Manager	Matthew Potter
Construction Environmental Manager	John O'Connor
Security Manager	Jeremy Dawes
Principal Designer Representative	Kevin Roberts
Occupational Health Lead	Charlie Lineham
Occupational Hygienist	Eric Ball

See organisational chart for Align: 1MCo5-ALJ-HR-CHT-C001-000001.

8.3 Responsibilities

8.3.1 The Project Director shall be responsible for:

- The overall direction, coordination, implementation, execution, control and completion of the Project, including the achievement of the Contract's KPIs.
- Leadership of the Health, Safety and Environment agenda through a consistent and energised approach.
- Leading the Quality agenda ensuring a "right first time" approach to the way we work.
- Leadership of the Project through establishing a capable leadership team who manage all aspects of the Project scope.
- Leadership of the Project by working effectively with HS2's organisation to ensure that all stakeholders are effectively managed.
- Ensuring that the Joint Venture is well managed and an open and respectful relationship exists between all the organisations.
- Ensuring communication in the Project, with HS2 and the other contractors is consistent and relevant.
- Ensuring clarity between Project scope, budget and overall deliverables.
- Ensuring that the Project is appropriately resourced.
- Maximising innovation, collaboration and efficiency.

8.3.2 The Director of Health & Safety shall monitor external developments and legislative requirements and translate these into Project Health and Safety Policy to ensure continued compliance and industry leadership and shall:

- Ensure these align with HS2, BYTP, VF and SRM policies.
- Develop, implement and maintain the Health & Safety Management System in all respects to ensure risks to third parties / contractors / stakeholders are understood and controlled.
- Collaborate with other Contractors to ensure creation of coherent and integrated policies and systems that avoid unnecessary duplication and dissipation of effort and to have better achievements.
- Provide advice on interpretation and application of the Health & Safety Management System and include provision of relevant training, briefings and information.
- Review the Scheme Design and issue comments in order to improve the health & safety during the site works.
- Manage the programme of audits of the Health & Safety Management Systems and ensure follow up and close out of actions where required.
- Co-ordinate and collate information required for the production of scheduled Health & Safety reports and identify trends and early warnings of possible issues and recommend appropriate action.
- Be accountable for effectiveness of the team and develop and ensure capabilities to support the function of the business.
- Be responsible for creating and communicating the Health & Safety and Contract Review vision and targets.
- Lead major incident /accident investigation.

8.3.3 Design Director

The roles and responsibilities of those in the design teams are laid out in the Design Management Plan, so the responsibilities of other people can be found there (1MCo5-ALJ-DS-PLN-C001-000001).

The Design Director has the following responsibilities:

- Accountable for ensuring the engineering scheme design is specified, delivered and maintained in compliance with HS2 requirements, including W1100 Health and Safety and W1150 Occupational Health.
- Principal point of contact regarding design for the project team with the Align-D board, Align, HS2 and stakeholders.
- Contributes to the development of Align-D's strategy across all areas of the project, challenging assumptions and decision-making as appropriate and providing analysis and guidance on all activities, plans, targets and project drivers.
- Build, lead, manage and motivate the Align-D project team. Take the lead role in directing all engineering activities and where appropriate prioritises activities in line with project needs.
- Develop and continually review the scheme design to ensure all project deliverables are aligned with HS2 requirements including the Health Hazard Evaluation and Mitigation Plans and information for the Health and Safety File.

8.3.4 Occupational Health Lead

The OH Manager is an integral part of the Health and Safety Team, supporting strategy development and implementation.

Specific responsibilities include but are not limited to:

- Review health information and statistics from design reviews, fitness for work outcomes, health surveillance grouped data, Health Impact Frequency Rates, wellbeing outcomes etc. to assist in the identification of health needs and health assessment activity.
- With the assistance of the wider occupational health team, create and update HMP, HHEMP and HRAs using all available health data such as; information on worker health outcomes and demographics of the workforce and community.
- Creation of a monthly report on occupational health outcomes and information for Health and Safety Performance Indicators (HSPIs).
- Participation in the Health Leadership Forum (HLF).
- Manage service provision and ensure that outcomes are delivered in line with agreements and standards.

8.3.5 Temporary Works Coordinator

Refer to the Temporary Works Management Plan (1MCo5-ALJ-CL-PLN-C001-000018).

8.4 Management System Arrangements

The ALIGN H&S Management System has been developed to demonstrate compliance with OHSAS 18001. This will be developed into an integrated management system with Quality and Environmental Management prior to Stage 2.

The implementation of the Code of Construction Practice is described in the Environmental Management Plan.

8.5 Health and Safety Advice

The Principal Contractor has a team of qualified health and safety professionals, currently made up of the Director of Health & Safety, Health & Safety Manager, Occupational Health Manager and a Health & Safety Advisor. Confirmation of competence for the roles is assured by ALIGN requesting and holding copies of the individuals' relevant professional qualifications as part of the on-boarding process.

General requirements for contractor Health and Safety Professionals are given in HS2's "Supply Chain Health and Safety Standard" [2].

During Stage One of the Main Civil Works, while there is limited construction work taking place on site, the H&S team will be small but will grow, as the scope of work increases into Stage Two, into a team of professionals that provides advice and support to all work activities within sector C1.

All contractors working on HS2 will have experienced and qualified health and safety professionals to provide competent advice to their management and workforce. Contractors (together with their sub-contractors) must agree with ALIGN how many contractor health and safety professionals are required on site, taking into consideration the nature of the work and the number of separate work faces.

All health and safety professionals employed in sector C1 will be required to attend HS2 training called Active Safety Leadership. This training will enable all Health & Safety professionals to deliver a consistent safety message across the sector as well as ALIGN specific health and safety training as required.

8.6 Assurance activities

8.6.1 Principal Contractor's Monitoring

Individual elements of the ALIGN's OH&S Management System implemented on the project, e.g. plans, procedures and policies, will be reviewed and assessed at least every six months in order to ensure that arrangements are continually improved.

A schedule of planned worksite inspections is developed in order to monitor OH&S performance in accordance with ALIGN's Contract Quality Plan (1MCo5-ALJ-QY-PLN-Coo1-000001). The schedule will include all health and safety inspections and safety engagement walks carried out by the Principal Contractor's site team including Construction, Site Operations and the Health and Safety Team. The frequency of worksite inspections will be in line with the hours worked on the site and will increase with the risk level of activities on the site. The assessment of level of risk is carried out using a system called Indice de Vigilance developed by Bouygues to ensure each risk area receives the appropriate level of attention. Quality, Environmental and security assurance activities are described in the respective Management Plan.

As a minimum weekly site team Health, Safety and Environmental inspections will take place by an appointed member of the site delivery team. In addition to a monthly Health, Safety and Environmental review by a member of the Health and Safety Team.

The schedule will be communicated to all persons required to undertake an inspection. The Project Director and Director of Health and Safety will ensure that performance against the schedule is continually monitored for compliance and will be regularly reviewed at meetings.

The ALIGN's Construction Team will carry out planned OH&S workplace inspections at locations and a frequency according to the OH&S risk associated with construction activities and also in response to any negative trends identified from the analysis of unsafe acts and conditions, incidents, safety inspections and safety engagement tours and construction surveillances. The Director of Health & Safety will maintain a schedule of planned inspections and monitor that inspections are being carried out to plan. The frequency of inspections will be continuously reviewed by the Director of Health & Safety in line with the risk profile and the results of other monitoring activities both by the Principal Contractor and other Contractors.

Copies of OH&S audits carried out on the project by external / client bodies shall be reviewed and maintained by the Principal Contractor's Health & Safety Team.

Senior ALIGN directors and managers based off-site will visit site on a monthly basis and undertake formal site OH&S tours. They will be accompanied by the most senior member of the Principal Contractor's team on site. Following the tour, the visiting Senior Manager will carry out a de-brief with ALIGN site management to discuss the observations and agree any corrective actions required. A written report will be made of the visit to ensure any actions are assigned and closed out.

Actions identified from inspections/tours will be addressed at the time of inspection/tour where possible and added to the project lessons learnt and incident management database, prioritised and regularly monitored to ensure they are closed out. The Site Management Team will develop improvement plans for any areas that require longer term improvements to the systems in use, with the support of the Health and Safety Team.

8.6.2 Contractors' Monitoring

Contractors' Managers, Supervisors (working and non-working foremen) and Health and Safety Advisors are responsible for ensuring each day that all work areas are safe before, during and after work activities. All Contractors are required to operate their own formal recorded inspection regime in parallel with that of the Principal Contractor.

Formal health and safety inspections - led by their Managers/Supervisors and supported by their health and safety professionals - will be carried out by all Contractors on a weekly basis as a minimum. A formal record of the inspections will be forwarded to the ALIGN Site Team and the Health and Safety team for review, collating and trending purposes.

8.6.3 Health and Safety Performance Index

ALIGN Director for Health and Safety will report to HS2 against certain measures for calculation of a Health and Safety Performance Index consisting of a Safety Performance Index and Health Performance Index upon such information being made available to ALIGN. All contractors will cooperate with ALIGN in the measurement of health and safety performance as and when reasonably requested.

8.6.4 Statistical Reporting

General requirements for statistical reporting are given in HS2's "Supply Chain Health and Safety Standard".

ALIGN will measure both leading and lagging OH&S Key Performance Indicators (KPIs). The key H&S statistical data that will be collected can be found in Appendix 2.

8.6.5 Review of the Construction Phase Plan

The Project Director will ensure that this Construction Phase Plan is reviewed and updated on a regular basis to ensure that:

- The objectives and requirements of the Construction Phase Plan are still valid and are being met.
- Future construction activities are reviewed and any necessary amendments to ALIGN's OH&S management processes or standards are put in place before the relevant work begins.
- OH&S management processes are reviewed to ensure continuing suitability and effectiveness.

This review will be done at least every 6 month or earlier if there is a significant change.

8.7 Communication and Consultation

It is recognised that effective, continuous communication is a key element of an effective safety culture, ensuring that everyone involved with the HS2 project is kept informed.

ALIGN will use the Communications Management Plan (1MCo5-ALJ-HR-PLN-Coo1-000005) to manage OH&S-related communications for the project.

Throughout the duration of the project, ALIGN will ensure there is regular communication with the workforce, providing information about the progress of the construction works, milestones achieved, upcoming challenges, safety incidents and information about local activities. This information will be shared via newsletters, Team Briefs led by senior management, poster campaigns and digital media. Site teams will be briefed regularly by site management and will also take part in safety stand-downs arranged at times around key construction dates, such as the start of a new element of the works, returning from Christmas break. All personnel will take part in a quarterly session as part of ALIGN's behavioural change programme.

Safety climate surveys will be carried out at least annually.

The ALIGN communications team will be responsible for all external communications relating to the HS2 project with the local community and external stakeholders.

Any interactions with the media must be directed through the communications team to HS2.

General requirements for Contractor workforce consultation are given in HS2's "Supply Chain Health and Safety Standard".

8.7.1 Regular Co-ordination Meetings

Coordination meetings will be held regularly with all Contractors on site to ensure that there is a clear understanding of forthcoming scheduled work activities and areas where there will be possible interface issues such as clashes and adjacency hazards. These meetings include:

- Daily coordination meeting (including Black Hat Meetings)
- Weekly coordination meetings with other PC at South Portal
- Weekly progress meeting
- Weekly integration meeting

The meetings will be chaired by a representative of the ALIGN Construction Team, and will include site supervision, contractors management and health and safety advisors.

8.7.2 Contract/Progress Meetings

ALIGN will hold scheduled contract/progress meetings with each Contractor which will include an in depth discussion of health and safety performance, any current issues and a review of progress in closing out any health and safety-related actions. These meetings will be attended by the management representatives of the Principal Contractor and Contractor, and the Principal Contractor's and the Contractor's Health and Safety Advisers.

Contractual communication between ALIGN and its contractors will be through CEMAR, although this channel is not yet in place. It will be in place before the end of Stage 1.

8.7.3 Safety, Security Health and Environment Leadership Team (SSHELT)

ALIGN has set up a monthly SSHELT meeting with the following remit:

The C1 SSHELT is a high-level meeting where Safety, Security, Health and Environmental strategy and significant matters for Sector C1 are discussed in the C1 senior leadership team attended by ALIGN and HS2. The meeting is the opportunity to review:

- Develop and feedback on ALIGN policies and strategies in
- HS2's 7 Focus Areas of safety
- ALIGN's 4w's Strategy
- HS2's Health Measures
- Security – Cyber and physical
- Environmental Management
- Highlight C1 activity outputs from related working groups with HS2
- Share best practice and lessons learned
- Review KPIs / HSPIs
- Promote communication between key stakeholders and disciplines – high-level knowledge share

8.7.4 Tool Box Talks

All Contractors working on site are expected to carry out regular toolbox talks (at least weekly) proactively with their work force to provide 'refresher training' on topics that are relevant to their current / future work activities.

In addition, contractors will deliver specific briefings created by the Principal Contractor to highlight issues that have been identified during site inspections or audits; or learning that has been established from the investigation of recent incidents.

Records of attendance and the information communicated will be retained by the Contractor organisation delivering the talks and issued to the ALIGN Health and Safety Team upon a weekly basis.

8.7.5 Non-English Speaking Workers

All Contractors working on the project must recognise the challenge that will be faced when communicating with workers for whom English is not their first language. This is particularly important when ensuring that safety-related instructions are being transmitted.

At the completion of the ALIGN site induction all attendees are required to complete a basic understanding question and answer session- this will allow confirmation of individuals understanding of basic English communication.

Where team members who do not have English as their first language, there shall always be a supervisor present with an individual, team or gang that can give instruction and translate where appropriate. The ratio of team members who do not have English as their first language to a supervisor with the ability to effectively communicate in English (verbal/spoken) shall never exceed 4:1.

8.7.6 Exchange of Design Information/Design Changes

Exchanges of design information will be carried out as part of the project design review meetings and be formally recorded. All design changes will be handled according to project procedures. Design changes will be managed in accordance with the HS2 Change Management Procedure (HS2-HS2-PC-PRO-000-000054). Each time the construction method is changed, the risk assessment must be reviewed along with the method statement. Any changes made to the method statement and/or risk assessment must then be briefed to all staff involved in the activity and others potentially affected by the changes.

Refer to HS2 Pre-Construction Information Pack (P1C1-HS2-PC-SCH-000-000001) section 5.2, which explains how HS2 discharges its duties as Principal Designer.

The design risk register shall be regularly reviewed by the designers to identify risks which may need to be elevated into the strategic or project risk registers and to monitor progress with mitigating design risks previously identified.

The provision of design information and the design risk register is detailed in the ALIGN-D Design Management Plan (1MCo5-ALJ-DS-PLN-Coo1-000001).

Specific information such as residual risk information will be included on the construction drawings in SHE Boxes and/or in the BIM model representation.

8.7.7 Selection and Control of Contractors and Suppliers

It is vital that the supply chain is fully engaged in ALIGN's Behavioural Plan Programme in order to achieve our aim of securing a safe working environment where everyone has the right to return home free from injury and ill health effects every day. In particular, ALIGN will ensure that:

- The occupational health and safety performance of suppliers and Contractors is key in the selection process.
- The competence of suppliers and contractors (particularly in relation to CDM) is key in the selection process.
- That the highest standards of occupational health and safety are specified in the procurement process.
- Selected suppliers and contractors are engaged early in the OH&S programme.

Contractor procurement will be carried out by the procurement team following ALIGN procedures. A health and safety questionnaire is issued to all potential Contractors. Completed questionnaires will be evaluated by the ALIGN Health & Safety Team, to ensure that Contractors meet the minimum OH&S competence requirements prior to being awarded any contract on the project in accordance with the ALIGN procedure.

Contractors must have arrangements in place that are satisfactory to ALIGN to ensure that any further subcontractors used are equally competent.

Contractor / Supply Chain on Site Kick-Off Meeting and Readiness Review

A contractor / Supply Chain On Site Kick-Off meeting will be held prior to the commencement of the construction activities of each Contractor company or prior to any significant change in risks in accordance with ALIGN Procedures.

Contractor Review of the Construction Phase Plan

Each contractor appointed to undertake construction work on the project will be provided with a copy of the latest revision of this Construction Phase Plan prior to the On Site Kick-Off Meeting.

Within 7 days of receipt, the Contractor's Senior Representative will review this Construction Phase Plan and shall inform the Principal Contractor if any of the Contractor's work activities create key health and safety risks that have not been adequately covered in this plan.

Contractor's Health and Safety Management Plan

Each Contractor is required to develop their own Health and Safety Management Plan that states how they intend to meet within their organisation and their sub-Contractors the requirements set out in this Construction Phase (Health and Safety) Plan; which nominates their safety-related appointments; and sets out the responsibilities of their site organisation. These plans will be reviewed and accepted by ALIGN as Principal Contractor before Contractors may commence work on site.

Management of delivery

Progress meetings will be held weekly with at least the Early Works Manager and the Contractor's Project Manager in attendance. An agenda for these meetings will include an assessment of the project's progress against the Accepted Programme; Technical issues; Health & Safety; Environment; Security etc.

Progress of the Early Works will be part of the PMU (Project Manager's Update) Report each month and will be reported on in the same way as the rest of contract 1MCo5. This will include commercial and other aspects.

Lessons learnt

The results of investigations into accidents, high potential near misses or any other significant experience gained on the project will be shared, initially with the project team and HS2, through HSSE Lessons Learnt posters or Alerts, depending on the severity and significance of the information. Following discussion with HS2 the lessons will then be posted on the Horace database and shared with the other JV's and wider industry if relevant. All ALIGN lessons learnt will be held in a database for ease of reference.

8.8 Co-operation and Coordination

ALIGN will liaise with HS2 Ltd, through the Project Manager in a timely manner, regarding any significant health and safety issues which arise during the design and construction phase, which may have an impact on the project or HS2 Ltd.'s activities.

ALIGN will liaise with the Principle Designer through the Project Manager on design carried out during Stage 1 and throughout the construction phases. This includes design by specialist contractors, and its implication on the CPP. Weekly Early Works Coordination meetings include an agenda item for any issue that needs to be escalated to the Principal Designer (PD) and, equally, for HS2 as the PD to communicate to ALIGN as the Principal Contractor.

ALIGN's activities will have interfaces with a number of significant other parties including:

- HS2 Ltd as designers of the railway systems
- HS2 Ltd contract areas S2 & C2 (the shared boundaries)
- Other principal Contractors (CEK and SCS), where ALIGN will take the lead in managing the interfaces
- Enabling Works Contractor (Fusion JV)
- Highways England and Connect Plus
- Network Rail and Chiltern Railways
- Utility providers
- Canal and Rivers Trust and Inland Waterways Association
- Members of the public
- Local councils and highway authorities
- Parish councils
- Natural England
- Environment Agency
- Private property owners (and/or tenants)
- The emergency services
- Hillingdon Outdoor Activities Centre (HOAC)
- Affinity Water

Senior ALIGN representatives and suitable others are to attend HS2 Ltd forums and meetings relating to H&S. These meetings include:

- HS2 SSHELT
- C1 SSHELT
- PC Forum
- Health Leadership Forum

8.8.1 Undertakings and Assurances

The requirements for Undertakings and Assurances (U&A) are varied across all construction disciplines. As such, the 'Plan for Compliance (CCP)' for each U&A is specific to that commitment. In the case that a physical constraint on site is stated by the U&A, it is important that operatives on site are fully aware of the constraint and Methods are planned accordingly. In most cases, this is dealt with by including details of the constraint onto Site Establishment Plans so that the constraint is clearly identified. Supplementary to that, necessary activities to guarantee the constraint is not breached, are included in the site RAMS documentation. These are communicated to the site teams via normal site procedure including 'toolbox talks'.

Assurance is carried out via weekly progress meetings and via physical auditing by the Align Requirements Engineer.

8.8.2 For the Early works

At a project level ALIGN coordinates its access to site through HS2 and the Site Tracker procedure on a weekly basis for the following week. ALIGN's contractors requiring access to any additional areas of land need to contact the ALIGN Project Manager and the Land Access Coordinator.

There is a weekly coordination meeting between ALIGN and Fusion, attended by ALIGN's Project Manager, Fusion's equivalent person responsible for the work in sector C1 and any other relevant people by exception.

There is a weekly progress meeting with the employer, HS2, attended by the ALIGN Project Manager, Site Manager, Health and Safety Manager and others to be agreed. From a health and safety point of view, this meeting enables the project to communicate any changes to risks, either identified by the client, which need to be cascaded or identified by the contractor and requiring escalation and sharing with others. The meeting is also used for the planning for upcoming activities and the hazards and risks associated with them.

In conjunction with HS2, ALIGN will arrange regular coordination meetings with other Principal Contractors working in nearby areas. Such other contractors may include Fusion, carrying out the Enabling Works, Osborne, for the M25 accesses, SCS for the adjacent sector S2 and others, as identified.

8.9 Competence

General requirements for competency of Contractors and workers are given in ALIGN's "H37 VW UK Accepted Record Schemes and Cards" which reflects the Construction Leadership Council (CLC) approved construction card schemes.

ALIGN fully acknowledges its duties under CDM and the duty placed upon Contractors to ensure any individual they employ or appoint to work on a construction site has the skills, knowledge, training and experience to carry out the work they will be employed to do in a way that secures the health and safety of themselves and others.

The minimum standard of health and safety training for all site personnel is a valid CSCS card or a Partner Card Scheme containing the CSCS logo. For Partner Scheme information please refer to appendix 7 for a full list of recognised Partner Schemes.

The card held by individuals must be reflective of the task/works to which they are appointed to undertake.

Exception to the above requirement in relation to each employee holding a valid CSCS is in relation to Security Personnel not employed to undertake any form of construction related activity. Where security personnel are undertaking general security guard duties they will hold a valid Security Industry Authority(SIA) card as a minimum which will be presented at the ALIGN on boarding process and display by the individual whilst at work.

Security guards with additional duties for example Traffic Marshal/Traffic Management duties will also possess a valid Site Access Traffic Marshal certificate of achievement (CLOCS)

In order to make sure that the security personnel is made aware of the risk related to construction the following measure will be implemented

- All the security personnel will follow the Site Workers induction
- The security personnel operation will be covered by a Site Specifics site
- As other personnel they will attend to the TBT
- A representative of the security company will attend to the daily coordination meeting ("Black Hat Meeting")

In addition, security personnel will not be allowed to access in any case to a live working area if not escorted by the relevant person

In addition to individuals holding a valid CSCS or a Partner Card Scheme containing the CSCS logo, individuals may need to hold supplementary training evidence to operate specific machinery, plant and equipment. For example, a steel erector required to use a MEWP would hold a valid CSCS card in addition to an IPAF card. In relation to accepted plant and equipment cards please refer to Appendix 7.

In accordance with the HS2 Supply Chain Health and Safety Standard the accepted standard of training for mobile plant operators is CPCS (BLUE) or NPORS (with CSCS hologram) or a Partner Card Scheme containing the CSCS logo. In relation to vehicle banksman training, the CLOCS Vehicle Marshal qualification is the recognised minimum standard of training required for vehicle marshalling upon site.

Any plant/equipment operator in possession of a Red CPCS Card (Trained Operative) will be subject to the contractor appointing a named competent supervisor to monitor works under the direction of the said operator. In addition evidence that the trade related NVQ has commenced will be provided by the contractor prior to Red CPCS card holders being permitted to operate any item of plant or equipment upon any ALIGN site/premises. The aforementioned is in addition to the completion of the employers competence statement which forms part of the ALIGN induction process.

In relation to Vehicle Banksman training, the CLOCS Site Access Traffic Marshal qualification is the recognised minimum standard of training required for vehicle marshalling upon site, and the use of STOP WORKS BOARDS between the public highway and the construction site entrance.

Operatives/Supervisors engaged in the installation, removal and maintenance of traffic management equipment upon a highway, shall have successfully completed the appropriate LANTRA module for the works to which they are undertaking. An operative holding a trainee card will be fully supervised by a competent member of the team holding a fully endorsed LANTRA qualification. Operatives/Supervisors engaged to undertake street works shall have successfully completed the appropriate New Roads and Street Works (NRSWA) module - appropriate to the works they are undertaking.

First Aiders - each contractor will provide sufficient trained first aid personnel for works under their direction. For construction related first aiders the required level of training will be the 3 Day First Aid at Work qualification.

All Contractors coming to site will map their roles to the Competence and Qualifications framework that will be used to verify that persons working on site are appropriately qualified and competent. Completion of an agreed matrix is a pre-requisite of mobilisation to site.

ALIGN ensures that all employees, visitors conducting more than three visits to HS2 Phase One working areas, subcontractors and suppliers and others within the working areas are in possession of a valid Construction Skills Certification Scheme (CSCS) card or a Partner Card Scheme containing the CSCS logo.

All contractors will be required to provide proof of training for their employees 72Hours in advance of attending the ALIGN site induction along with a fully completed ALIGN Employee/Employer declaration of Competence, Validate application form and completion of the Park Health Pre Employment Health Questionnaire, which is to be brought to the ALIGN induction by each individual. All requested documentation and supporting evidence, with the expectation of the Park Health Pre Employment Questionnaire, are to be forwarded to the ALIGN Induction e-mail address - Inductions@alignjv.com at least 72 hours in advance of the preferred induction date. Upon receipt of the fully completed forms and supporting documentation, the date, time and location of the next available induction will be advised by return e-mail. Pre Commencement Health Forms are **NOT** to be issued to ALIGN by e-mail, but brought to the induction by each individual and inserted into a sealed envelope for issue to Park health for review.

In exceptional circumstances, and where a particular trade is not covered by a recognised card scheme, the contractor, in conjunction with the Principal Contractors Health and Safety Team, will review and agree upon an alternative acceptable standard of training.

Working over or near water, minimum training standards:

ALIGN will ensure the following minimum standards of training for personnel working on, adjacent or above water where full collective protection cannot be provided and maintained or works that are within 3m of the water course:

- DEFRA Level 1- Water Awareness for supervisors, managers and survey works not involving works upon the water but in close proximity.
- DEFRA Level 2- Flood and swift water responder for personnel and supervisors/managers for physical works being undertaken on or above water.
- DEFRA Level 3-Water Rescue technician for personnel employed and appointed to provide rescue from water

ALIGN acknowledge the supply chain may have higher standards than the above and in such instances the higher standard will be accepted by prior agreement with ALIGN Health and Safety Team.

All Tunnel Personnel:

A Minimum of the City and Guilds 6151 level 2 qualification will be required for tunnel entry and tunnelling activities and this must be supplemented by the CITB and British Tunnelling Society (BTS) endorsed Tunnel Safety Training Scheme (TSTS) tunnelling qualification.

Confined Space work personnel:

In all cases, the confined space qualifications will be City and Guilds 6150 and to the appropriate risk and competence level.

Asbestos Training:

Personnel required to undertake work or surveys where there is a foreseeable risk of ACMs being encountered must hold a valid UKATA approved Asbestos Awareness certificate which is renewed at periods not exceeding 12 months

8.9.1 Site Induction

One of the key elements of our health and safety programme at HS2 is our site induction. It underpins compliance with the requirements placed upon the Principal Contractor under the CDM regulations such as providing all workers with information relating to the project, configuration of the site, and details of emergency arrangements. It also sets out our Behavioural expectations - and defines what is different about our project. Anyone working on the project (regardless of location) will be required to attend the Project Introduction.

There are three types of induction relating to the construction phase as follows:

- Visitors induction
- Delivery driver's/Maintenance Engineers' induction
- Site Workers induction (including Security Personnel)

Refer to ALIGN Induction Procedure (1MCo5-ALJ-HS-PRO-Coo1-000003) for full details and requirements relating to the above types of induction.

Completion of the Site Workers' Induction is mandatory to obtain full and unescorted access to the Project and this will be recorded by the ALIGN On-Boarding Team. The Worker's Code of Conduct is covered during the Project Introduction where inductees agree to this showing their commitment to the ALIGN values and expectations whilst working in the local community.

The following are all pre-requisites for contractor's employees obtaining an ALIGN site access pass:

- Construction Skills Certification Scheme card - CSCS or a Partner Card Scheme containing the CSCS logo, suitable for their role on site. Where a specific card does not apply to an individual i.e.: students on work experience, ALIGN will ensure such persons are fully supervised at all times by an appointed competent supervisor.
- Checked against a standardised Competence and Qualification Matrix where individuals are registered as either being 'competent' or 'competent with a training plan' depending

on the Competencies and Qualifications linked to a specific role. This also applies to all Supervisors.

- Completion of a Pre Placement Health Questionnaire, with a hard copy brought to the induction by each individual and inserted into a sealed envelope for issue to Park Health for review.
- Completion of a Validate Access Application form
- Production of a current Safety Critical Medical if undertaking a safety critical role.
- Completion of Employee/Employer declaration and competence form.
- Successful completion of the ALIGN Site Induction (including Workers Code of Conduct).
- Successful completion of the Q&A section of the ALIGN Site Induction.
- Evidence of being briefed upon their employers Status A RAMS for works to which each individual is appointed to undertake.

Contractors are responsible for providing all their workforce including sub-contractors with a company induction and site orientation. Additionally, Safety Critical Medicals must be arranged by the employer and recorded by Park Health prior to safety critical activities commencing. (See Appendix 3 for a list of occupations requiring Safety Critical Medicals). This ensures that individuals understand the specific rules applicable to the work activities they will carry out and to reinforce the particular requirements and safety standards of ALIGN, including emergency arrangements. Contractors shall submit copies of their inductions/orientations to ALIGN for review and shall provide records of attendance to ALIGN upon a weekly basis.

In advance of the HS2 Safety Passport scheme being introduced across the HS2 Contracts, ALIGN will be adopting the HS2 Validate Card Scheme to control access to all ALIGN Construction Areas for ALL Construction related workers.

The Validate Card Scheme will be a mandatory requirement for all construction related workers to gain access to all ALIGN Construction Areas as of 1st January 2019. As part of the ALIGN pre-registration process, 72 Hours in advance of persons attending the ALIGN site induction the following information will be provided for each individual requiring an ALIGN site induction:

- Completed Employee/Employer declaration form
- Completed Pre placement health questionnaire, with hard copy to be brought to the site induction by each individual.
- Valid skill card
- Valid safety critical medical if applicable
- Completed Validate application form

Upon the above information being received and reviewed by ALIGN, confirmation of induction attendance will be confirmed to the employer along with an Induction date and time slot.

Upon the employer accepting the allocated date and time slot for an individual ALIGN will apply for each individuals Validate Card which will be available for collection following the ALIGN site induction.

Following the completion of the ALIGN induction process and issue of the Validate Card, access to any ALIGN Construction site will only be permitted upon presentation and issue of the Validate Card which will be held by security and returned to the individual upon departure.

Worker - A worker is someone who will be carrying out tasks that require a Risk Assessment, Permit to Work or a Work Authorisation (including Generic RAMS's).

A worker can be working anywhere on site and may not necessarily be carrying out construction work or 'manual' labour.

Visitors Non-Working - Those carrying out tasks that do not require a Risk Assessment, Permit to Work Order or a Work Authorisation (including Generic RAMS)

Delivery Driver/Service Engineer - Is someone who is delivering/ collecting product/material/goods on a HGV over 3.5tonnes to the project or undertaking plant and equipment maintenance.

8.9.2 Control of Persons Entering the Site

The general requirements for entry to site and authorisation for pedestrian access to the site are given in ALIGN's "Supply Chain Health and Safety Standard".

Authorisation for people to enter site will be controlled by a Site Access System called Validate for the main contract works. The intent of HS2 Limited as the system owner is to ensure that the individuals entering their sites are doing so safely and securely. The HS2 Site Access System will be used to help ensure that only those holding the correct competences are undertaking works.

The Validate system will hold information about the holder of the Validate card, including the date and type of induction received, CSCS or CPCS card type among other relevant information on competence of the individual. The individual's Validate card can be checked at any time by someone with a suitable card reader in order to verify the person's competence to carry out a certain activity.

Until the H&S passport system is fully functional, and in addition to recording information on the Validate system, individuals will receive a helmet sticker indicating that the wearer has completed the site induction.

In addition each person will be identified by the means of a name stick up on their helmet.

All people (including delivery drivers and Visitors) display a pass (as a means of identity) or have such identification readily available for purposes of identification. ALIGN will refuse entry to or remove from the Working Areas anyone who refuses to supply a means of identity. The Validate card and access to the site may be withdrawn at any time.

HS2 is currently developing (in conjunction with Network Rail and Transport for London) a new H&S Passport system. The H&S Passport will be introduced to sector C1 (ALIGN's contract) when it is rolled out to HS2 (likely to be early 2019).

8.9.3 Access to Working Areas

All working areas will have a controlled access point where employees and Visitors pass through and sign in (manual or electronic) on entry. All people attending the Working Areas (for example a delivery driver) are advised of the relevant Working Area risks, control measures and ALIGN's Working Area rules in the form of The Delivery Drivers Briefing issued to them.

No one under the age of 16 is permitted in Working Areas. In special circumstances (for example an organised visit from a local school) where people of 16 years or under are admitted to the Working Areas (as Visitors only), ALIGN will carry out a Risk Assessment documenting the controls necessary to safeguard their safety, health and wellbeing. This also applies to passengers in vehicles making delivery to or visiting the Working Area for any other reasons.

8.10 Welfare Facilities

General requirements for Contractor-provided welfare facilities are given in HS2's "Supply Chain Health and Safety Standard" and in Schedule 2 of the CDM Regulations.

Contractors must ensure that there are adequate welfare facilities i.e. canteens, rest facilities, changing and drying rooms, washing and toilet facilities and a supply of fresh drinking water available to their workforce and any subcontractors. The minimum requirement for the welfare should be as per Workplace (Health, Safety and Welfare) Regulations 1992 and associated ACoP.

Where contractors provide welfare facilities as part of their contract the proposed facilities will be agreed in advance with ALIGN Management to ensure a satisfactory standard of provisions are provided. Separate facilities will be provided for male and female personnel at all work locations.

The standard of welfare to be provided at each location for the works will be described in the RAMS for each location. As a minimum it must meet the requirements laid out in section 4.4 of the Supply Chain Health and Safety Standard, ALIGN guidance note "Welfare Facilities and Site Set-up" and will be agreed with HS2 in advance of mobilisation.

Contractors must ensure that their facilities are maintained to a high standard of cleanliness and repair and the consumable items are replenished at least daily or when required. Records of inspections will be maintained.

Particular attention will be paid to ensure that there are properly constructed footpaths, steps, and staircases with non-slip treads and handrails to allow safe access to these buildings. The exposed edges of all steps will be highlighted to prevent trips.

All persons must use the welfare facilities provided to change. Changing in car parks is not permitted.

8.11 Notification, Reporting and Investigation of Incidents

The terms “incident, accident and near-miss” are as defined in OHSAS 18001. Refer to the Site Emergency Preparedness and Response Plan [1MCo5-ALJ-HS-PLN-C001-000032] for the details of those with specific responsibilities in the case of emergency situations on site and the processes to follow.

Generic requirements for the notification, reporting and investigation of incidents are given in HS2’s “Supply Chain Health and Safety Standard”, ALIGN’s Crisis Management Plan. Further details on procedures to be followed are contained in ALIGN procedure “Accident/Incident Investigation and Reporting”.

All work-related incidents (i.e. accidents resulting in personal injury or ill health, or near-misses with the potential for personal injury or ill health) must be formally reported to HS2 by the responsible ALIGN manager by completing an Incident Report in HORACE (HS2’s continuous improvement and incident management tool).

Each incident shall also be reviewed to determine if the actual or potential consequence meets the criteria for a Reportable Incident under the RIDDOR Regulations.

Onward internal notifications to ALIGN Management and externally to others will be made in accordance with ALIGN’s procedure “Accident/Incident Investigation and Reporting”. Incidents with significant actual or potential consequences will also be notified to the JV partner companies via their own chain of reporting.

In addition to its own procedures for reporting any accident to an employee or member of the public, Contractors shall provide a copy of statutory notification Form No 2508, if applicable, and a copy of its investigation report, to ALIGN’s Health and Safety Department. All specified injuries, occupational diseases and dangerous occurrences as defined by RIDDOR (Reporting of Injuries, Disease and Dangerous Occurrences Regulations, 2013) shall be reported immediately to the Principal Contractor’s Health and Safety Manager.

Note: Any RIDDOR notifications to the Health and Safety Executive will be coordinated by ALIGN’s Health & Safety team, with submission of the Statutory Notification (F2508) being made by the relevant employer following consultation with the ALIGN H&S Team prior to any such submission.

All incidents will be subject to an initial report being issued within 24 Hours followed by a completed report normally within 7 days of the recorded accident/incident.

8.11.1 Reporting of Hazards (Unsafe Acts & Unsafe Conditions)

Generic requirements for the reporting of observed Unsafe Behaviours and Unsafe Conditions are given in HS2’s “Supply Chain Health and Safety Standard” [2].

A key part of proactively preventing incidents is the active participation of all site personnel in the identification of unsafe behaviours and unsafe conditions; taking ownership of them; taking actions to correct them (which includes intervening in an imminent danger situation);

and then reporting their observations and actions taken i.e. "See it, Sort it, Report it, Learn from it".

A simple Workplace Observation Report (paper format) is provided by the Principal Contractor. This Workplace Observation form must be used to report health, safety, welfare and environmental observations, both excellent behaviours/conditions and unsafe behaviours or unsafe conditions, and what was done to correct/control the situation at the time. ALIGN will provide "suggestion boxes" for paper workplace observation reports to be posted.

All personnel on site are encouraged to raise any health and safety issue, either formally or informally, with their respective supervisor or safety representative. Regular feedback on any further corrective actions taken to address the concern raised will be provided to employees either individually by their supervisor or via toolbox talks and briefings.

Persons involved in, or seeing a near-miss or accident, must tell their Supervisor or Manager so that they can raise an Incident Report.

8.11.2 Confidential Reporting of Serious Concerns

Staff should normally initially raise concerns through their Line Supervisor / Manager or direct report if working on a sub-contract basis. If the Health & Safety issue is serious and likely to place staff in imminent danger they shall IMMEDIATELY stop work until the issue is resolved. Raising concerns whether minor or major, can be done without fear of victimisation or detriment. Concerns may be raised either verbally or in writing. If it is felt that the matter cannot be raised with the immediate Line Supervisor / Manager the concerns can be escalated through the Management Team and where necessary, direct to the Project Manager.

8.11.3 Enforcement Notices

Generic requirements for Contractors to report any enforcement notices are given in HS2's "Supply Chain Health and Safety Standard".

Contractors shall inform ALIGN of any regulatory interest that takes place during their works at HS2 or elsewhere.

8.12 Risk Assessments and Method Statements

8.12.1 Preparation, Review and Acceptance of Risk Assessments & Method Statements (RAMS)

The aim of the ALIGN assessment system is to break the work activity down into systematic steps from start to completion, at each step identifying the hazards and where possible eliminating the hazards as far as reasonably practicable. Where elimination is not practicable the controls to be used must be identified. The use of PPE is only to be considered as a last resort, as indicated in the hierarchy of controls. See ALIGN Procedure "Risk Assessment Procedure - 1MCo5-ALJ-HS-PRO-C001-000005".

8.12.2 RAMS for Construction Activities

All Contractors shall prepare and submit to the Principal Contractor prior to their On-Site Kick-Off Meeting a list of the risk assessments and method statements (RAMS) that they will

prepare for their initial works to be carried out on site. This information shall continue to be provided to the Principal Contractor's Site Team during regular progress meetings and used by the Principal Contractor to maintain a register of all RAMS on site.

All contractors shall ensure that Pre-Construction information, including residual risks from the Designer Risk Register is incorporated into the health and safety plan and in the RAMS documents.

Risk assessments and method statements shall be prepared and submitted to the Principal Contractor at least 14 days (or as stated in the contract) prior to the commencement of all surveys, construction, commissioning and maintenance tasks involving physical work on site.

All RAMS shall be reviewed by the Contractor's Health and Safety Professional prior to submission to the Principal Contractor. As part of this review, the Contractor's Health and Safety Professional will identify any safety critical activities/high risks that require hold points be included in the method statement and within the Inspection and Test Plan, requiring notification by the Contractor to the Principal Contractor before proceeding.

All RAMS shall be prepared, reviewed and accepted in accordance with ALIGN Project Work Instruction "Procedure for Method Statements" In summary, this work instruction requires the following:

- Both Contractors and ALIGN will prepare RAMS for all their surveys, construction, commissioning or maintenance work tasks on site.
- Contractors will use their own risk assessment process/templates/risk matrices to prepare their risk assessments for their work tasks.
- ALIGN will use its own risk assessment process/templates/risk matrices to prepare its risk assessments for its work tasks.
- By prior agreement, a peer review will be carried out for all high risk activities.
- All RAMS will be issued as "Fit for Construction" to the Principal Contractor.
- All RAMS will be assessed by the Principal Contractor against risk criteria.
- All RAMS will be reviewed by the Principal Contractor.
- All RAMS, will be formally accepted by the Principal Contractor as Status A – "Accepted" prior to work commencing.
- All RAMS will be itemised on a RAMS register created and maintained by the Principal Contractor.
- Work Authorisations and Permits to Work will only be issued if a Status A – "Accepted" RAMS exists on the RAMS Register.
- All RAMS will be cancelled once the task is completed or if more than 6 months old.
- All current RAMs will be subject to a weekly review between ALIGN and the Contractor to ensure scheduled works are covered within the current revision and foreseeable risks are identified and adequately controlled where such risks cannot be mitigated.

Using a standard method statement template helps reduce the amount of generic information or duplication of content that can mask important task/location specific information. It also aids the authors and reviewers since information is presented in a

consistent order to a common standard so speeding up the preparation and review process and helping to identify any omissions.

For complex tasks and at the Principal Contractor's discretion, the ALIGN Project Team will arrange and chair a workshop involving the Contractor and the Principal Contractor's nominated RAMS reviewers and other stakeholders to carry out a team based risk assessment (TBRA). A TBRA reduces abortive work and delays to the review process by ensuring that RAMS authors and the allocated reviewers have understood the proposed construction method; identified all significant hazards; selected and agreed in principle suitable and practical control measures; and identified any safety critical hold points.

Note: Only in exceptional circumstances (such as to remedy a safety issue or for emergency work that could not have been reasonably foreseen) will RAMS be accepted if issued less than 14 days in advance.

Method statements shall include suitable and sufficient information on the following:

- Task title, start date and duration.
- Description of the works (a step-by-step process for completing an activity).
- Identification of drawings relevant to the task.
- Identification of any permits and consents required.
- Resources required (competent people, Equipment, Plant and Materials).
- Equipment, Plant and Materials to be used and any assessments needed or authorisations for use.
- Assessment of significant health, safety, security and environmental risks and impacts for all tasks and activities.
- Control measures and Safe Systems of Work.
- Personal Protective Equipment.
- Emergency arrangements and initial incident response.
- Temporary amended systems.
- Monitoring and compliance.
- Risk Assessment.
- Specific Equipment required to complete the task.
- Relevant Undertakings and Assurances (U&A's) relevant to the location and planned activity.

Method statements shall also:

- Be expressed in an easily understandable manner using simple language.
- Define clearly and concisely the safe system of work developed by the contractor which integrates the health, safety, environmental, security and ecological control measures identified by risk assessment into the steps required to complete the task.
- Result in effective communication of this safe system of work from a contractor to his workforce via his Supervisors.
- Identify hold points and/or high risk steps in the task at which interventions/monitoring should take place.

Risk assessments and method statements (RAMS) will only be prepared by person(s) who have received suitable risk assessment training and suitable task knowledge and experience.

Method statements are sometimes lengthy documents which describe either a major activity or a generic task which may be carried out at different times or locations. In these cases, Task Briefing Sheets or similar should be used to break down the key elements of work described in the method statement for different work groups/teams.

Clear communication of accepted work methods and controls defined in risk assessments and method statements must take place prior to work commencing via a pre-job brief. This is to be carried out using a task briefing sheet delivered by a manager or supervisor. The delivery method must be closely matched to the competence and language skills of the recipients to ensure that there is a true understanding of the work method. Briefings will be recorded on signed task briefing sheets in addition to a section being included to question the workforce upon their understanding of the approved RAMs and in particular the identified control measures. For subcontracted work, all Contractors are to retain copies of their lower tier contractor briefing record and a manager is to attend a sufficient number of briefings to ensure the standard and level of information given is acceptable.

Task briefings are in addition to the Contractor's Daily Activity Briefings with his team covering specific hazards such as severe weather conditions, work interfaces and lessons learnt from the preceding shift.

The Principal Contractor will identify health and safety interventions/monitoring to confirm compliance with RAMS through planned inspections by the Principal Contractor's Team.

Where activities cannot be carried out as per the RAMS, the contractor's Supervisor must ensure that the work ceases until the activity is reassessed and either an addendum to the RAMS is prepared or the RAMS is revised, reviewed by the Contractor Safety Professional, accepted by the Principal Contractor and re-issued.

8.12.3 Work Authorisation / Permits for Work

The Principal Contractor will operate a Work Authorisation and Permit for Work System to control all construction, maintenance and commissioning activities in sector C1 in accordance with "ALIGN Permits to Work Procedure – 1MCo5-ALJ-HS-PRO-Coo1-000006", and the generic requirements given in HS2's "Supply Chain Health and Safety Standard".

All construction work must have a work authorisation issued by ALIGN and all high risk work will be subject to the Permit to Work System to include:

- Permit to Break Ground
- Hot Works Permit
- Permit to work
- Permit to cross utilities/work in close proximity of overhead services.
- Permit to remove vegetation
- Permit to work over/near water
- Permit to use ladders/Step Ladder
- Permit for Confined Space Works
- Permit to work on/test electrical equipment

8.12.4 Point of Work Risk Assessment (POWRA)

All Principal Contractor and contractor personnel are required to complete a Point of Work Risk Assessment (POWRA) at the workplace prior to commencing any new physical work activity or when conditions change and at least daily in line with the STAR Principles:

- Stop
- Think
- Act
- Review

The POWRA is the final step in the work planning process. A POWRA is a process to confirm that the work planning process (including risk assessments and method statements) has adequately identified all hazards and control measures to carry out a task safely; or to identify unanticipated hazards and the additional controls required; or deal with changes in conditions/personnel etc. It also ensures that the hazards and controls are fully understood and accepted by the team carrying out the task.

The process of developing a POWRA before starting or continuing a work task helps ensure proper OH&S planning by ensuring the team “stop and think”. By involving each team member in task planning before work begins they are encouraged to think about the hazards involved and controls required and so own the safety process. Developing the POWRA also identifies resources (tools, equipment, materials, and personnel) to ensure their availability at the point of work which improves efficiency and also helps prevent the taking of short-cuts or improvisation which often leads to accidents.

If, during a task, additional hazards are identified e.g. conditions change (ground conditions or severe weather/environmental conditions), personnel change, or a deficiency in the plan is discovered, work shall stop. The current POWRA is then modified or a new POWRA created to eliminate or control the additional hazards.

If the task changes or the POWRA process identifies the need to change the agreed work methodology defined in the method statement, then work cannot continue until a revised method statement has been prepared and approved as above.

A POWRA may not be used to justify deviating from the method of working defined in an approved method statement.

The POWRA is developed on site at the point of work (not in the office) by the team assigned to perform the work, guided by the supervisor (working or non-working foreman) who leads the discussion.

Any objection to the content of the POWRA should be raised with the contractor’s Supervisor for discussion and resolution. If the objection cannot be resolved after escalating the

objection to the Contractor's Site Management team then refusal to sign the POWRA constitutes refusal to work on the grounds of safety.

In summary, POWRAs are used by competent workers (already briefed by their Supervisor on the RAMS) to check that before starting or continuing a task:

- They understand what it is they're about to do
- They know how they're going to do it safely
- There are enough people to do the work
- They have all tools and equipment they need
- There are no previously unanticipated hazards
- Conditions have not changed

Each contractor work crew under the leadership of a working foreman (ganger/charge hand/supervisor) must complete a POWRA. Where the team identifies hazards that are not adequately controlled, then they will ensure that the work does not start/continue until the hazard has been suitably controlled. This may involve addressing the hazard at source or reporting to their supervisor.

Within 2 hours of the task commencing, non-working contractor foremen/supervisors shall visit each work location and challenge work crews on their understanding of the work activity (and agreed control measures) and shall review and counter-sign the POWRA. The Point of Work Risk Assessment does not replace the legal requirement for a risk assessment.

8.13 Personal Protective Equipment

The Contractor shall:

- Provide Personal Protective Equipment (PPE) in accordance with any risk assessment and mitigation control
- Procure PPE from companies who are members of the Registered Safety Suppliers or similar scheme, such as SEDEX (ethical trading)
- Ensure orange PPE, with long sleeve tops, is worn for all worksite activity
- Develop a process for regular inspection, training and replacement of PPE.

Individual's employers shall provide the following PPE, which will be worn by all individuals when entering any designated PPE zones, within the worksite:

- Safety helmets with 4 point chin strap, that comply with BS EN 13688:2013 Protective Clothing – General Requirements, (although an exception may be made for Sikhs wearing turbans, who do not wish to wear a safety helmet on religious grounds, where risk assessment determines that a turban provides adequate protection from the risk of head injury). Please note; only purpose made hard hat liners are permitted to be worn beneath any hard hat, (hoodies/bobble hats are not permitted).
- Hi-visibility tops (Long sleeved t-shirts or polo shirts, vests, jackets) orange with reflective tape, which complies with EN ISO 20471 (formally EN ISO 471:2003) class 2, in addition to long sleeves - no bare chests
- Hi-visibility long trousers (working trousers and over trousers) orange with reflective tape compliant with EN ISO 471:2003 class 2

- Safety footwear, which complies with BS EN ISO 20345:2011 Personal Protective Equipment, and also includes:
 - Support to the ankle
 - Covered toe protection
 - Mid-sole protection
 - Metatarsal protection (Wellington Boots to be S5 as a minimum)

Rigger boots are prohibited

- Eye protection:
 - That conforms to BS EN166:2002 1.F Personal Eye Protection Specifications
 - Prescription safety glasses for individuals who require them.
- Hand protection that conforms to BS EN388:2003 intermediate design (Minimum cut 3)
- Hearing protection that conforms to BS EN 352.

Task specific PPE will be subject to risk assessment and may include the following:

- Personal head lamps for persons working at night and ATEX rated intrinsically safe equipment for those working in tunnelling and below ground environments
- For persons exposed to working in inclement weather conditions, Hi visibility (EN ISO 20471 class 2) protective equipment which complies with EN 343:2003 Class 3:3
- For persons working under a hot works permit and/or in close proximity to buried services and electrical installations, all clothing must be Hi visibility flame resistant, antistatic and arc flash resistant PPE which complies with EN ISO 11612 (FR), EN1149 (AS), EN ISO 20471 (HV) and IEC61482 (ARC)
- For persons working in close proximity to concrete application shall wear Tyvek coveralls, together with items identified in the risk assessment
- For persons engaged in Works Over or Near Water, without collective protection in place, suitable PPE including life-jackets conforming to EN399 (ISO 12402-2) rated at 275 Newtons
- In confined space and tunnel area, all PPE including lights and gas monitors shall be intrinsically safe.
- Where RPE is required the minimum standard to be provided with be FFP3 with all wearers of RPE being subject to Face Fit with evidence of passing the face fit being provided.

All High-Visibility PPE must:

- Carry the Principal Contractor's logos
- Conform to the Employer's corporate branding requirements
- Be submitted to the Project Manager for acceptance.

Generic requirements for personal protective equipment are given in HS2's "Supply Chain Health and Safety Standard" and "Personal Protective Equipment"

Additional or specialist PPE may be required for particular tasks as determined by risk assessment e.g. personal flotation devices; metatarsal protection flame retardant coveralls when welding or when excavating in the vicinity of buried electrical services; goggles for

grinding, rotary disc cutting or scabbling; respiratory protection; fall restraint/arrest systems; and safety wellington boots if working in water is unavoidable.

A point of work risk assessment (POWRA) signed off by a non-working foreman, supervisor or manager may be used to justify not wearing an item of standard PPE if wearing that item of PPE creates a significant additional hazard e.g. an operative can be given permission to not wear light eye protection if working in the rain well away from any work activities that could result in an eye injury.

Hand protection is to be worn in all construction, workshop, and maintenance areas at all times. Hand protection chosen must be suitable for the task and provide an appropriate level of abrasion, cut, tear and puncture protection. Specific hand protection requirements and glove types must be identified by risk assessment and defined in each method statement.

Mirrored light eye protection is not permitted. Tinted light eye protection may only be worn outdoors in sunny conditions unless welding.

Respiratory Protection will be a minimum of FFP3, JSP 'Click to Check' half mask, with evidence of Face fit being provided for each individual expected to wear such PPE.

All PPE provided must be maintained in a good condition and provisions made for its replacement if it becomes damaged, its storage and cleaning as required. Contractors are expected to maintain uniform PPE standards across its employees. All PPE must comply with the HS2 Specification for Personal Protective Equipment and indicate the actual sub-contractors name as employed by ALIGN and not that of any sub-sub-contractor.

Contractors shall ensure that their personnel are suitably protected against the weather.

To help reduce the effects of hot weather working personnel working in the construction areas shall be provided with either:

- One-piece light weight coveralls.
- Two-piece coveralls (top and bottom) which let in some air.
- A combination of overall bottoms with a long sleeved high visibility polo/rugby shirt with collar.

Contractors shall ensure compliance by its employees and/or visitors. ALIGN reserves the right to instruct the removal from site of anyone not complying with such a mandated requirement.

ALIGN reserves the right to direct contractors to change/replace personal protective equipment if they determine that it is worn out, unsuitable or inadequate for its proposed use.

Contractors must have suitable arrangements in place to ensure the cleanliness of any items of personal protective equipment provided to its employees e.g. a laundry facility for overalls, high visibility clothing and other work wear.

Vehicle Marshals, and Vehicle Banksman will wear yellow safety helmets to enable them to be easily recognised by reversing vehicle drivers and mobile construction equipment operators.

Slinger/Signallers (lifting operations) will wear orange safety helmets to enable them to be easily recognised by crane operators and lift supervisors.

Suitably coloured safety helmets (identifying the name of the person and their company) will be worn by all persons in the construction areas on site to aid in the identification of key roles as follows:

Role	Safety Helmet Colour
Contractor Working and Non-Working Foremen/Supervisors (gangers/charge hands)	Black
Medical Staff	Green
Trainees, Apprentices or Young Persons (16 to 18 years old)	Red
All Visitors not carrying out physical work	Blue
All other contractor personnel/operatives	White
All Banksmen, Vehicle Marshals	Yellow
Slings/Signallers	Orange

It is essential that the crucially important role of the foreman/supervisor (ganger/charge hand) in achieving our AIM High philosophy is recognised and their status is not undermined. By wearing a black coloured safety helmet, the supervisor is clearly distinguishable from his work crew and can be easily identified if any person wishes to ask a question; challenge the crew over an observed unsafe act or condition; or give positive reinforcement/feedback.

Working and non-working foremen who are also first aiders will be identified by a sticker applied to their safety helmet.

Drivers of 4x4s on site must wear (or have available) a full set of mandatory PPE; passengers must also wear PPE if they intend to exit the vehicle on site.

Escorted visitors who are not carrying out physical work may enter live construction areas on foot to view the works from designated viewing areas or at the discretion of their escort but must wear at all times as a minimum the following PPE:

- Blue safety helmet with chin strap
- Light eye protection
- Long sleeved high visibility jacket (orange)
- Long trousers (orange)
- Gloves
- Safety Wellington boots (or lace up safety boots if preferred) to S5 specification.
- Hearing protection

8.14 Site Rules

To help communicate and provide a focus on our key health and safety requirements, ALIGN has two types of site rules at HS2.

ALIGN has introduced these rules with the goal of preserving life, if any of these cannot be met, work must immediately stop.

- Life Saving Rules General
- Life Saving Rules - Site and Field Safety

8.14.1 Life Saving Rules - General Safety

	I never use a mobile phone while driving
	I never work or drive under the influence of alcohol or drugs
	I always wear my seat belt and respect speed limits

8.14.2 Life Saving Rules – Site and Field Safety

	I never walk or stand under a suspended load
	I always wear a life jacket when working near water in the absence of collective protection
	I always maintain a safe distance from moving equipment
	I always protect myself against falling from height and protect others from falling objects
	I always obtain authorisation, atmosphere control and supervision before entering a confined space

8.14.3 Data protection.

Align adopts the principles set out in the Data Protection Act 2018 and the General Data Protection Regulation and puts appropriate measures in place to make sure that personal data collected through the induction process and/or engagement on the works is stored securely and protected against unauthorised or unlawful processing, accidental loss, destruction or damage. Personal data is retained only for its specified and legitimate purpose and is kept only for the period of time necessary.

Occupational Health records are collected, stored, processed and retained by Park Health & Safety in accordance with their Data Protection Policy and Privacy Notice.

9 Emergency Arrangements

9.1 Emergency Procedures

ALIGN has developed an Emergency Preparedness Plan which has been developed to cover the early construction activities. Contractor emergency arrangement compliance is a requirement for the approval of work authorisations and hold points. It should be noted that individuals from the Principal Contractor and contractor organisations are assigned roles, activities and responsibilities to provide safe response to the following types of incidents:

- Construction
- Security
- Environmental
- Fire
- Severe weather

The emergency arrangements will be regularly tested and practised. Emergency exercises will be conducted to demonstrate the adequacy of the emergency arrangements, and all staff, contractors and visitors involved with the project shall participate in the exercise programme as required. Contractors will need to prepare and execute a programme of drills and exercises to cover the risks identified in their working areas.

9.1.1 Accounting for People working

ALIGN will be using the site access system known as Validate to confirm the competence and authorisation of people to work on site. This system will be developed with technology to enable site management to account for people working on site, using passive systems to identify the location of individual workers particularly in tunnelling.

9.1.2 Fire Safety

Generic requirements to ensure fire safety are given in HS2's "Supply Chain Health and Safety Standard".

Fire Safety Risk Assessments and the "Sector C1 Fire Safety Plan", that detail the site-wide fire safety arrangements for sector C1 have been developed by ALIGN using procedure "Provide Fire Prevention and Fire Safety Arrangements", the Regulatory Reform (Fire Safety) Order 2005, the Fire Protection Association Fire Protection on Construction Sites, and the Joint Code of Practice on the Protection from Fire of Construction Sites and Buildings Undergoing Construction. For the early works) the fire plan as developed by ALIGN will apply to the Compound Managed directly by ALIGN. With remote work locations being subject to a fire risk assessment being undertaken by the appointed contractors engaged at the said work location. Align will review the compliance of those Fire Risk Assessments

The locations of external fire assembly points will be displayed on site notice boards. These must be brief during site induction and updated regularly at tool box talks. Contractors will need to identify suitable fire assembly points for their working areas.

9.1.3 Emergency Medical/First Aid Response

ALIGN will operate a coordinated Emergency Medical/First Aid Response to all persons on site utilising resources available on site from ALIGN and all Contractors.

First Aid kits shall be provided by the contractors in all their site offices, mobile construction equipment with enclosed cabs and in site vehicles where determined within specific first aid risk assessments. Emergency stations with first aid kits and eye wash facilities will be provided by the ALIGN at various locations around the main compound. First aid facilities will be clearly identified with signage.

Requests for First Aid/Medical Emergency response to anywhere on the ALIGN site will be made by contacting the ALIGN emergency number, or appointed first aiders mobile phone numbers, which will be provided at induction.

Posters displaying first aiders and fire marshals will be displayed upon site notice boards for ease of identification of suitably trained personnel.

9.1.4 Security Lockdown

In the event of unauthorised intruders on site, the site location may be directed to go into a "Security Lockdown". This procedure will follow the site-specific security plan with the aim of getting all personnel into a safe place as quickly as safely possible. Training on these procedures will be provided in due course. Refer to the ALIGN Incident Management Plan, Appendix A which describes the Bronze, Silver, Gold approach to levels of incident and threat.

10 Controlling Significant Site Risks

10.1 Dealing with Utility Services

ALIGN is clear that buried services present one of the most significant hazards on the project. For this reason, there is a team dedicated to obtaining and providing utilities information and the associated permits for any occasion when our operations require the ground to be penetrated or broken. ALIGN's procedures for dealing with buried services during the Ground Investigation works are described in H14 - Safe Working Procedure for Working in the Vicinity of Underground Services. In addition all works requiring the breaking of ground for any purpose will be subject to full controls as defined within the ALIGN Permit Procedure.

10.1.1 Avoiding overhead services

In conjunction with our Permitting procedures, ALIGN requires all contractors to follow the HSE's guidance document "Avoiding danger from overhead power lines" Guidance Note GS6, which can be downloaded for free from the HSE website.

In addition a Permit to pass under/work near overhead power lines will be implemented upon the project

10.2 Temporary Works

During the early Works in Stage 1 it is not anticipated to undertake any structural temporary works as defined in BS5975: 2008. However, there will be temporary elements of construction required to be designed for the early works and these will include, but not be limited to:

- Hoardings for compounds
- Haul roads and amendments to accesses
- Protection to buried services
- "Goalposts" and other demarcation of safe crossing points under overhead services
- Pile and working platforms

All construction elements of a temporary nature will be designed in accordance with the BYUK Temporary Works Management Procedure, which will be reviewed and developed further, if necessary, to comply with "Technical Standard – Temporary Works" (HS2-HS2-CV-STD-000-000005). The BYUK Temporary Works Management Procedure is already compliant with BS5975: 2008 and it will be checked against:

- BSI PAS 8811 'Temporary works – Major infrastructure client procedures – Code of practice'
- BSI PAS 8812 'Temporary works – Application of European Standards in design – Guide'

10.3 Working at Height

A work place is 'at height' if a person could be injured falling from it, even if it is at or below ground level (i.e. excavations and service pits). RAMS for work activities involving work at height shall define the safe system of work that will ensure that:

- All work at height is properly planned and organised
- All work at height takes account of weather conditions that could endanger health and safety
- Those involved in work at height are trained and competent
- The place where work at height is done is safe
- Equipment for work at height is appropriately inspected
- The risks from fragile surfaces are properly controlled
- The risks from falling objects (including tools) are properly controlled. Tools and equipment used at height which represent a foreseeable risk of being dropped/falling from height will be tethered.
- Persons working at height are in visual and verbal contact

Work at height access equipment must be suitable for the task and of good condition.

Safety nets, air bags or other similar devices shall not be used as the only means of fall protection.

An emergency rescue plan shall be in place whenever fall restraint, fall arrest or safety nets are used; and a safe means of rescuing promptly a fallen worker or incapacitated worker shall be available at all times. The emergency rescue plan can be defined in the RAMS or in a separate plan and must include the provision of rescue resources for work at height, either for rescue of an incapacitated person by others, or self-rescue of an individual without additional assistance.

If any Work at Height is found to be necessary, it will be subject to the ALIGN Work at Height Procedure.

10.4 Scaffolding

Not required during Early Works.

10.5 Ladders

Stepladders shall not be used as working platforms unless a risk assessment has demonstrated that mobile towers or podium scaffolds cannot be reasonably used for the proposed work. Step ladders may only be considered for non-complex activities of short duration. Short duration but repetitive task e.g. cable tying or fitting brackets, multiple lamp changing or bracket installation on pipe runs would be seen as one activity and as such the risk assessment would have to justify why a platform had not been provided for the extended works. Any step ladder used must have collective protection built into the apparatus.

Ladders are only permitted when they form part of a proprietary access system with full collective protection being maintained.

10.6 Mobile Elevating Working Platforms (MEWPs)

All works involving the use of a Mobile Elevated Work Platform will be undertaken in full compliance with the ALIGN Mobile Elevated Work Platform Procedure in addition to task specific Status A RAMs being in place.

10.7 Prohibited types of plant/equipment

The following items of plant and equipment are prohibited upon any ALIGN works site:

- Ladders unless they form part of a propriety access system
- Traditional forward tipping dumpers
- 125mm and smaller angle grinders/cutters
- The use of Acetylene is prohibited upon all ALIGN work sites

10.8 Pressurised systems

The use of any pressurised air/gas or similar product is only to be considered where no safer alternative method is available (ie: use of water)

Prior to the use of any pressurised air/gas or similar product where the above criteria cannot be met the following must be in place:

- An approved fully robust operating procedure covering all elements of the planned activity
- A schematic diagram of all pressurised systems and labels on control valves showing positive and negative pressure on and off
- Pressurised systems must have a minimum of 1 fail safe valve preventing the build-up of excess pressure within any system
- The pressurised product is provided at the lowest possible pressure to complete the scheduled task, achieved through but not limited to low pressure supply (de rated bottles) and low rated regulators upon bottles/canisters
- All components and fittings within the pressurised system must be in receipt of valid evidence of strength and integrity with a suitable safety factor applied
- Pressurised systems are subject to a pressure test / leak test" in a controlled environment and a pre use check witnessed by a supervisor with records of such tests recorded
- All persons involved in the planned activities must be competent and have received suitable training in the use of and handling of compressed air/gas
- All supervisors/managers involved in the planned activities must be competent and have received suitable training in the use of and handling of compressed air/gas

10.9 Traffic Management and Segregation of Vehicles and Pedestrians

ALIGN Procedure "Specific Risk Assessment - Traffic and Pedestrian Management" will be utilised to develop the site traffic management plan at each location.

The site traffic management plan, and in particular the routes to be taken, will change frequently throughout the duration of the project and details of traffic routes will be posted on site notice boards, and discussed at daily coordination meetings.

The first consideration will be to provide separate traffic and pedestrian routes with physical barriers in place to prevent people/traffic interfaces.

ALIGN have implemented a coloured zonal guidance including the use of coloured barriers to identify transit and traffic routes as follows:

White Barriers - A non-construction (white) zone - Areas where no construction control (e.g. PPE) is required. For example designated welfare and private vehicle parking areas.

Green Barriers - A protected and segregated environment for navigating around or through a red or amber (restricted or controlled) zone. For example- designated pedestrian routes on and through sites.

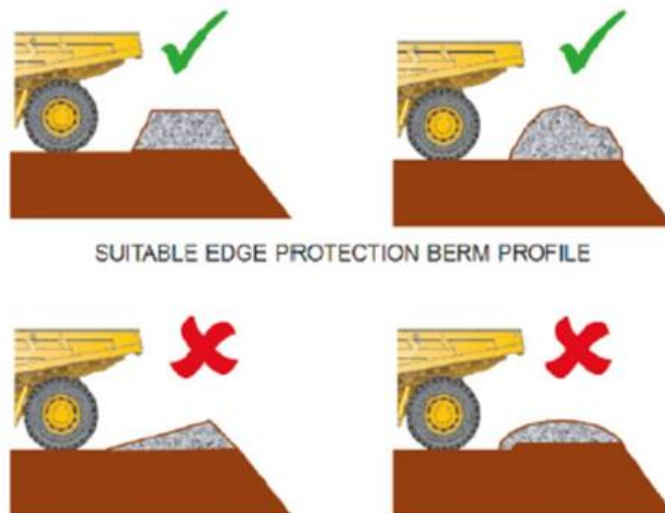
Amber barriers - A controlled (amber) zone – Access control systems in place to allow only those trained to operate in or persons authorised by the areas Black Hat. For example - pilling operations, plant activities where access is required under controlled conditions.

Red Barriers - A restricted (red) zone – Unambiguous physical demarcation and no pedestrians allowed. For bat/badger exclusion zones, contaminated land, confined spaces where no access is required, work areas where remote controlled plant is in use.

Transitional points will be identified with red hooped barriers with sprung loaded gates and zonal signage being clearly displayed.

Protection of work platforms and haul roads to prevent over run:

Work areas and traffic routes will be adequately protected to leading edges as to prevent mobile plant and equipment from over running work platforms and haul roads. Upon raised working platforms and haul roads where there is a foreseeable risk of overrun such works areas and transit routes will be protected by a bund or similar to a minimum of half the height of the largest wheel or tracked items of plant/equipment being operated within the zone as to prevent over run. See examples below:



10.10 Site Vehicle Drivers

The ALIGN Local Traffic Management Plan (LTMP) (1MCo5-ALJ-EV-PLN-Coo1-000003) describes ALIGN's response to High Speed 2 Phase One: Route-Wide Traffic Management Plan 2nd Edition (HS2-HS2-CL-PLN-000-000002) and has been prepared specifically in relation to the Early Works.

With the exception of HGVs and mobile construction equipment, all vehicles in designated parking areas in construction areas or in site car parks shall be reverse parked. Reverse into your parking space when you arrive so it is easier for you, and safer for pedestrians, to drive out forwards when you leave.

10.11 Delivery Vehicle Drivers

Generic requirements for delivery vehicle drivers are given in HS2's "Supply Chain Health and Safety Standard". In addition to the general requirements, there is a specific undertaking and assurance regarding Construction Logistics and Cyclist Safety (CLOCS), with which all delivery vehicles and drivers are required to comply. The details of this undertaking and assurance can be found in Appendix 6.

10.12 Unloading/Loading of Materials and Equipment from Vehicles

Unloading and loading of materials and equipment from vehicles for the ground investigation works shall be specifically covered in the RAMS for the work. However, all vehicle operators and those responsible for vehicle loading and unloading must be trained and competent in loading and unloading their vehicles and trailers. The Driver and Vehicle Standards Agency has produced the useful guide on this subject which can be found at the following link:

<https://www.gov.uk/government/publications/load-securing-vehicle-operator-guidance/load-securing-vehicle-operator-guidance>

Where there is a need to access the rear of any trailer then edge protection will be provided along with a safe means of access and egress.

For any lifting refer to ALIGN lifting Procedure [1MCo5-ALJ-HS-PRO-C001-000031](#)

10.13 Loading/Offloading of Mobile Construction Equipment from Vehicles

Generic requirements for controlling the health and safety risks from loading/offloading vehicles are given in HS2's "Supply Chain Health and Safety Standard" and "Specific Risk Assessment - Workshop - Loading and Unloading Vehicles".

10.14 Poor Ground Conditions

During Early Works, contractors and others visiting the site locations should expect to encounter poor ground conditions, dependent on the location, season and prevailing weather conditions. The local ground conditions will be assessed as part of the point of work risk assessment (POWRA) and taken into account when planning a safe method of work, which will include assessment of the type of vehicle, track matting, alternative permitted routes etc. All of the above will be considered when deciding to proceed with the work at that time and location. On no account will the safety or health of people be put at risk due to poor ground conditions.

Early Works RAMS will include a generic method for extricating vehicles and other equipment that may have become bogged down in poor ground.

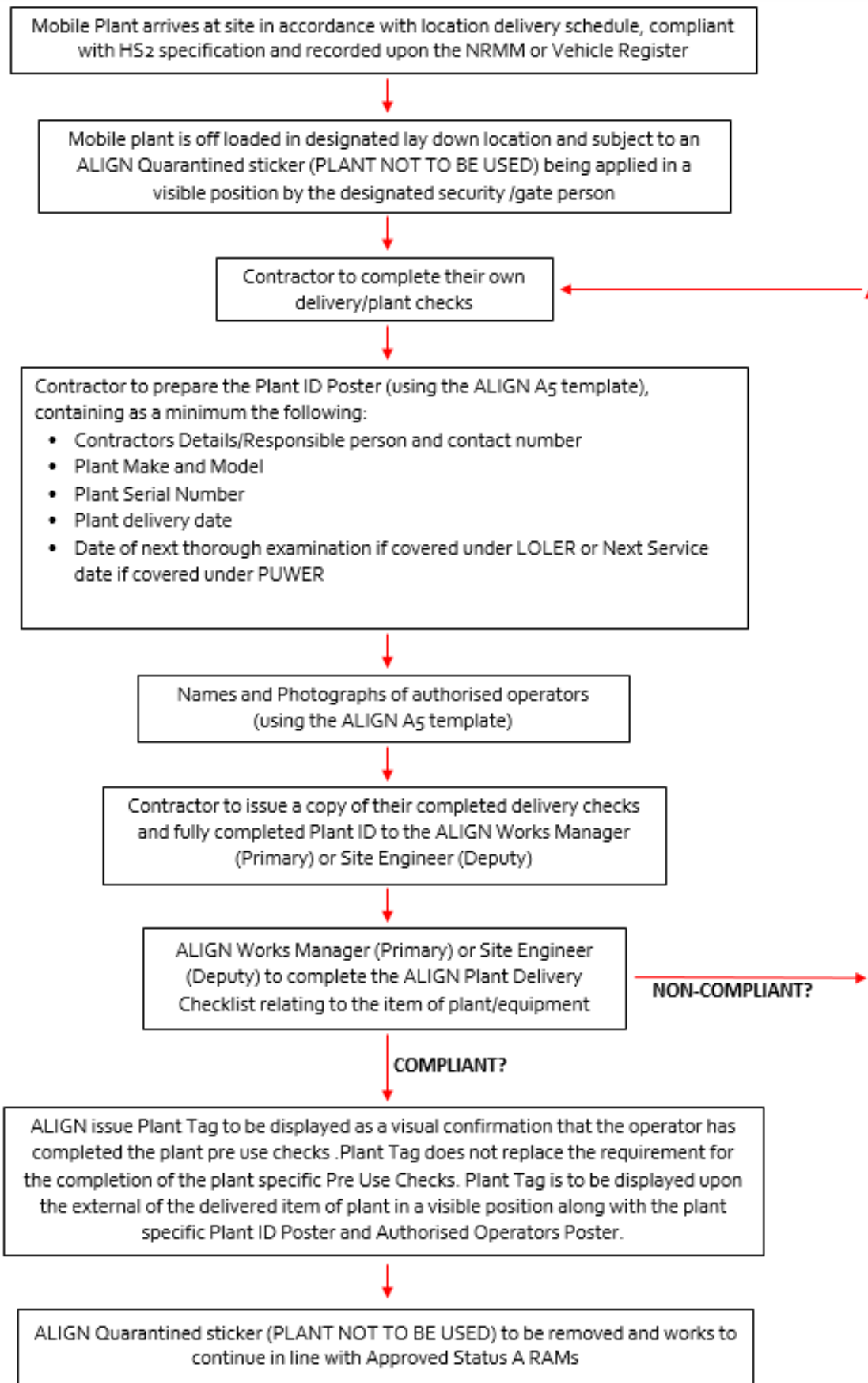
At any location where construction vehicles enter or leave the highway, the contractor shall ensure that mud and other debris is not deposited on the highway. This may require the use of a jet wash or similar method of removing mud etc. from the body and wheels of vehicles using the access. If the highway does become contaminated in such a way, the contractor shall ensure that it is cleaned immediately and at no point is left in a slippery or other condition that could become a hazard to passing motorists or other road users. The contractor will have an arrangement with a road sweeper to keep the carriageway clean as and when necessary.

At LTP 2 works location, ground enhancement will be undertaken with ground stabilisation works under the direction of Geobind.

10.15 Mobile Construction Equipment

A full record of mobile plant currently upon the project will be maintained by each appointed contractor for all plant under their control and that of their supply chain.

All mobile plant being delivered to any ALIGN site/premises will be subject to checks and verification in accordance with the Mobile Plant Delivery and Authorised Use Process, on the next page:



10.16 Other Plant and Equipment

Generic requirements for controlling health and safety risks from other plant and equipment are given in ALIGN's Supply Chain Health and Safety Standard.

In accordance with the HS2 Supply Chain Health and Safety Standard the accepted standard of training for mobile plant is CPCS or NPORS (with CSCS hologram) or a Partner Card Scheme containing the CSCS logo.

In relation to the undertaking of vehicle marshal duties upon a highway the required standard of training is LANTRA 12D.

To ensure clear communication links are maintained between mobile construction plant operators and persons who are employed to direct and monitor safety critical times of mobile plant, for example:

- Communication between cranes and slinger/signallers
- Communication between concrete pump operators and nozzle men
- Excavators/Dozers and appointed banksman
- Telehandlers and vehicle marshals

Contractors will employ a 2 way communication system such as Duplex Dect Comm11 to ensure continual communication between operators and safety critical personnel is maintained.

Restriction upon forward tipping dumpers:

The use of forward tipping dumpers is not permitted upon any ALIGN works site. Contractors will source alternatives such as:

- Hydrema - all round visibility dumper with cab
- Wacker Neuson - Dual View with swivel driver's seat cabbed dumper

Quick Hitch restriction upon HS2/ ALIGN sites is as follows:

All semi-automatic and First Generation Manual Hitch attachments are prohibited upon at all work locations. Only direct connection or fully automatic quick hitch attachments are permitted upon any ALIGN site.

Concrete and pump lines:

The primary method of cleaning out any concrete or grout line will be with the use of water. Contractors are to ensure they have a robust documented risk assessment for such tasks with the mandatory use of a ball (pig) catcher attachment and a means of constant communication between the pump operator and assistant.

However where there is no other practicable alternative to clean out such lines in the conventional method using water the use of compressed air is permissible with the following in place:

- Fully robust RAMs having been submitted and approved in advance
- Eye protection is replaced with a full face visor
- No personnel are within close proximity of the pipe line and in particular at the discharge end
- All flexible hoses are removed (does not apply to CFA ground hoses)
- Pipe line is secure and supported
- No bends within the final 15m of the pipeline
- Ball catcher attachment must be fitted to the discharge end
- The compressed air must be introduced gradually and not exceed 7 bar
- A competent assistant is appointed to assist the pump operator

Accommodating Adjacent Land Use:

Land Access is arranged by ALIGN on a permitting basis through HS2. No access is permitted to any land that is not:

- A public right of way
- Land covered by a current valid permit issued to ALIGN
- Land expressly identified as prohibited access in the contractor's permit to work issued by ALIGN

Landowners:

Landowners requiring access to their land through land occupied by ALIGN and its contractors will be accommodated as far as reasonably practicable and if safe to do so. The Landowner requiring access should contact the ALIGN Community Engagement Team in the first instance to raise a request. The request will be passed to the Construction Management Team and meetings held, if necessary, between the landowner and site team to plan the provision of access to the landowner. All further dealings with the landowner may be carried out directly by the site team but will be monitored by the Community Engagement Team. Access will only be granted to the landowner if it has been properly planned in advance, all risks associated with the interaction assessed and mitigated, and the site team briefed on the plan.

10.17 Work On or Near Fragile Materials

No fragile materials are expected to be encountered during the early works.

10.18 The Control of Lifting Operations

A single colour-coded tagging system will be adopted for all lifting accessories (including webbing slings, chain slings, and all types of shackles) on HS2 sector C1 owned by contractors or ALIGN that will be changed every six months. The colour scheme is shown below:

Orange	1 st JULY 2019	To	1st January 2020
White	1st January 2020	To	1st July 2020
Green	1st July 2020	To	1st January 2021
Yellow	1st January 2021	To	1st July 2021
Blue	1 st July 2021	To	1 st January 2022

Contractors shall display the current colour for lifting accessories on boards in their compounds.

Please note: Lifting accessories belonging to suppliers undertaking in frequent deliveries (one off deliveries) to ALIGN sites do not need to be identified with colour coded tags defined in ALIGN's Lifting Procedure. However, thorough examination certificates must be submitted with these accessories to support the specific lifting plan, the accessories must be identifiable and a recorded check undertaken prior to any lift commencing.

All certificate shall be maintained by each contractor, at the site location for inspection.

The ALIGN appointed "Lifting Supervisor" will review all lifting plans. All lifting operations as a minimum will be subject to an approved lifting plan and lifting risk assessment and a team briefing prompt sheet.

All lifting accessories are to be in receipt of valid thorough examination (every 6 months) in addition to being subject to daily pre use checks and weekly LOLER inspections.

In addition all lifting accessories will be tagged in accordance with the current colour coding as identified above

ALIGN prohibit the use of single use slings/lifting accessories. Whilst ALIGN encourage the use the use pre-slung loads all lifting accessories will be certified with evidence of certification (individual or batch) being provided.

10.19 Excavations

Generic requirements for controlling health and safety risks from excavations are given in HS2's "Supply Chain Health and Safety Standard" and "H14 Safe Working Procedure for Working in the Vicinity of Underground Services".

Contractors undertaking any works involving the breaking of ground will ensure a Cat B PAS 128 survey has been completed prior to works commencing, with the ALIGN Permit to Break Ground being used for all such works.

All formed excavations will be adequately protected to prevent persons and plant from falling within any formed excavation, with specific details to be identified within the Contractors RAMs.

ALIGN uses the following guidance notes to ensure safe excavation practices:

- ALIGN Excavations and Ground Support Systems
- ALIGN Safe use of Excavators

10.20 Confined Spaces

It is expected that there will be a need for working in confined spaces during the Early Works. All such works will be undertaken in accordance with ALIGN Confined Space Procedure with personnel required to enter a confined space being training to City and Guilds 6150 and to the appropriate risk and competence level.

10.21 Work Near Water Giving a Risk of Drowning

Generic requirements for work near water giving risk to drowning are given in HS2's "Supply Chain Health and Safety Standard"

ALIGN's detailed procedures and training requirements for personnel working within 3m of the water's edge, and for those working over water, are detailed within the ALIGN Work on or Over Water procedure - 1MCo5-ALJ-HS-PRO-Coo1-000012

10.22 Explosives and Cartridge tools

Contractors shall not bring onto the site explosives or explosive devices including cartridge operated tools without prior written consent of the Principal Contractor. Contractors shall be responsible for obtaining all necessary licences or permits for such items of equipment.

10.23 Housekeeping and Storage of Materials and Work Equipment

Generic requirements for housekeeping and storage of materials are given in HS2's "Supply Chain Health and Safety Standard"

Contractors will provide suitable storage areas for materials and work equipment without exception.

At all times safe access and egress around stored items must be maintained.

Particular attention must be given to storage of items and their stacking height to ensure that hazards from falling objects are removed.

All loose materials and components are to be stored in crates or stillage's as to reduce the potential of trip hazards.

10.24 Portable Electrical Equipment

Generic requirements for controlling health and safety risks from portable electrical equipment are given in HS2's "Supply Chain Health and Safety Standard".

All items of electrical equipment used upon the construction areas will be subject to 3 monthly PAT with current test information to be made available and items of equipment displaying current information.

10.25 Erection of Structures

During site mobilisation, there will be a requirement for contractors to establish site compounds. Contractors shall ensure that all lifting operations and work at height activities are carried out in accordance with HS2's "Supply Chain Health and Safety Standard" and "Work at Height"

10.26 Severe Weather Conditions

Effective arrangements to be detailed within Contractors RAMs to include but not limited to Risks Associated with Flooding and severe weather conditions.

Adverse Weather controls are detailed within the ALIGN Adverse Weather Procedure - 1MCo5-ALJ-HS-PRO-C001-000016, with the detailed control measures and precautions being applied to all works under the direction of ALIGN and appointed contractors.

10.27 Fire Hazards

Rubbish or other materials must not be allowed to accumulate to prevent potential fire hazards. In particular cabin heaters must be kept clear of clothing and other electrical items turned off at the end of each shift or when the premises are unattended.

All exit routes will be kept clear at all times and access to emergency exits maintained.

Fire hazards in the locations of intended works must be considered in the risk assessments and method statements. Contractors must assess each work location and implement appropriate fire precautions as appropriate.

Every site compound shall be subject to a specific Fire Risk Assessment every six months carried out by a competent person. Fire hazards will also be incorporated into weekly health and safety inspections and other monitoring of all offices and site compounds.

10.28 Hand Held Work Equipment

The contractor shall ensure that the usage of hand held work equipment including scythes, pneumatic breakers, shovels, saws; chain saws, wood chippers, core cutting equipment etc. are covered by appropriate risk assessments and method statements.

All hand tools will be subject to compliance with Provision and Use of Work Equipment Regulation 1998 (PUWER) including but not limited to pre use inspections.

The use of 125mm and angle grinders/cutters less than 125mm, are prohibited upon ALIGN projects.

10.29 Gas Monitoring

Contractors must ensure that appropriate gas monitoring is undertaken whilst site activities are being undertaken which require ground disturbance such as trial pits, boreholes, excavations etc.

This requirement shall be included as part of the risk assessments and method statements developed by contractors.

10.30 Interaction with the Public

All communications with the public must be done through the Communications Team and Community Engagement Team. The only exceptions to this are the Community Safety Champions and in the event of an emergency, which requires direct involvement with the public in order to save harm, injury or damage to property. The strategy for community engagement is laid out in the Community Engagement Plan.

10.31 Communications (mobile phones/radios) and audio equipment

For Early Works it is not practical to install a radio system so mobile phones will be main means of communication. Each location will be tested before work starts to ensure there is adequate mobile phone signal. If there is poor phone signal in the area, alternative means of communication will be set up before any work starts in that location. This may include a pager or satellite phone etc. Designated Heras fenced areas ("Phone Zones") will be created at each site where mobile phones will be used for operational communications or essential personal communications off site.

10.32 Sharp Objects including Barbed Wire and Thorns

All areas where personnel may come into contact with barbed wire or other sharp objects shall be assessed and suitable precautions put into place. Particular attention will be paid to the likelihood of fly tipping at or near ALIGN work areas. A point of work risk assessment will be undertaken for any instance of fly tipping that needs to be managed by ALIGN contractors. This will include the assessment of the presence of sharp objects, including hypodermic needles, blades of any kind and thorns.

This requirement shall be included as part of the risk assessments and method statements developed by the contractors

10.33 Demolition Works

None required during this stage of the works, however some demolition survey works will be undertaken (i.e. Asbestos Surveys and structural stability surveys)

10.34 Work in Colne Valley lakes and other bodies of water

The Colne Valley viaduct passes over a number of water courses and lakes and will necessitate a large amount of work in close/3m proximity to water and at height over water. Materials and

other resources will be delivered to viaduct pier locations using waterborne transport. Early Works will take place in these locations using jack-up barges or pontoons. Arrangements will be developed to be explicit on the standards expected, the local emergency arrangements, equipment, training, competency etc., before anyone is required to work in these areas. Protection of water bodies and water courses will be explained in the specific section of the appropriated RAMS as and when required.

10.35 De-watering

During the-Early Works it is expected that some pumping tests will be required but no de-watering as such. Pumping tests will be properly designed, planned and executed.

10.36 Piling

Test Pile/Load Pile works are scheduled for January 2019 with such works to be controlled in accordance with Approved Status A RAMs which will be made available to HS2 for review and comment in advance of such works commencing.

11 Occupational Health and Wellbeing

Refer to the ALIGN Health Management Plan (1MCo5-ALJ-HS-PLN-C001-000003 for details of how Health is managed at ALIGN JV.

11.1 Contaminated Land

- 11.1.1 It is likely that the scope of this Contract will involve access to, or intrusive investigation of, land which is classed as Contaminated. Sites have been classified for the purpose of H&S by HS2 Ltd in accordance with the Guidance for Safe Intrusive Activities on Contaminated or Potentially Contaminated Land (BDA, 2008). Refer to Section 4.14 for HS2 Ltd Phase One environmental statement volume 2: community forum area reports and map books for baseline information gathered for the Environmental Statement.

11.2 Asbestos Containing Materials

The asbestos risk for all works, including demolition and excavation in made ground, needs to be assessed and controlled by Contractors in accordance with the Control of Asbestos Regulations 2012 and the Approved Code of Practice.

Areas of potentially contaminated ground including ACM contamination are detailed upon Contam Composite Plan 1 dated 28/09/2018 and Contam Composite Plan 2 dated 28/09/2018.

Personnel involved in works or supervising works which could uncover potential ACMs will be in receipt of a valid asbestos awareness qualification (UKATA).

Contractor's RAMS will include arrangements for dealing with suspected ACMs during the course of their works, in addition to compliance with the ALIGN Unidentified Finds Procedure – 1MCo5-ALJ-HS-PRO-C001-000015 and referring to the Asbestos in Soils Procedure [Document no: 1MCo5-ALJ-HS-PRO-C001-000026]

11.3 Control and Use of Hazardous Substances

Any use of hazardous substances will be managed through the use of COSHH risk assessments based on the Material Safety Data Sheets from the manufacturers and an assessment of the way in which the substance will be used. Refer to ALIGN procedure COSHH, which describes the hierarchy of control to be applied when there is a potential need for a hazardous substance to be used on an ALIGN site.

COSHH storage will be identified on the Site Plan and records will be maintained at the site location.

11.4 UXO and UXB

The available UXB/UXO maps for the length of the route have been obtained and they indicate a low risk of encountering unexploded ordnance (Zetica 2012). UXB mapping is only at a regional scale and relates to bombs dropped, not bombs stored.

A map of the unexploded ordnances in Sector C1 with reference 0615-ZET-GT-REP-000-000001, to assist in identifying any areas of risk in the Zetica Report was used to support completion of a detailed risk assessment.

11.5 Manual Handling

Contractors must ensure that they comply with the Manual Handling Regulations by following this simple hierarchy:

- Avoid hazardous manual handling operations likely to cause injury so far as is reasonably practicable
- Assess any hazardous manual handling operations that cannot be avoided
- Reduce the risk of injury so far as is reasonably practicable by implementing appropriate controls

Manual handling must be a specific consideration in every risk assessment produced for activities on the project and a specific manual handling assessment produced by competent persons.

Appropriate controls should consider the use of mechanical means to reduce exposure to muscular skeletal disorders and repetitive strain injury including advancements in available technology.

Contractors must ensure that their personnel and those of their sub-contractors have received adequate Manual Handling training with evidence being provided to ALIGN if manual handling works are to be undertaken.

11.6 Noise and Vibration

ALIGN have developed a Noise Management Plan for the sites and these plans will be included in the EW EMP. Contractors undertaking work or services on site which may generate noise at or in excess of the lower exposure action value, shall provide a written assessment to ALIGN. The assessment shall detail precautions to reduce exposure to a safe

level both for its own employees as well as others who may be affected and health surveillance required. Minimum SNR Rating for hearing protection will be identified within contractors RAMS and enforced at the work location.

All tools, which emit sound pressure levels at or above 85 dBA are fitted with a blue (mandatory) "hearing protection required" sign. When such a tool is being operated, employees are instructed to wear hearing protection and must have adequate control measures put into place as detailed within the task specific Risk Assessment. Additional control measures may include, but not limited to mandatory hearing protection zones or exclusion areas. Consideration shall be given for the protection of the public in the areas where noisy works will be performed

ALIGN acknowledges its own responsibilities in regard of the above and will co-ordinate and inform contractors as appropriate on such matters.

ALIGN will undertake occupational and environmental noise surveys for demolition, dismantling, construction and commissioning activities to ensure personnel are not exposed to excessive noise levels and that any local or site constraints are not contravened.

Noisy evening and night time construction work will be avoided, in line with the relevant Planning Conditions.

Contractors shall only use hand tools and equipment selected to minimise exposure to noise and vibration if they are unable to eliminate the requirement to use them. The contractor will monitor the HAVS exposure levels and take the necessary steps to keep within the A8 daily Exposure Action Value of 2.5m/s². Where required additional personal monitoring will be conducted by the contractor to ascertain the suitability of the tools/equipment being used. These can then be substituted for those that emit less vibration.

For reference please note: Ground Investigation Noise Risk Assessment - 1MCo5-ALJ-EV-REP-C001-600001

11.7 Exposure to UV Radiation (from the sun)

Exposure to UV Radiation (From the Sun) will be managed with a cover up campaign being implemented along with personnel protection advice being available to the inherent risks associate with such exposure.

11.8 Substance Abuse

Everyone working on the ALIGN project shall comply with the ALIGN "Alcohol and Drug Policy".

Workers must come to work fit for duty and ensure that they are not intoxicated by alcohol and are free from the effects of any illegal drugs when they report for work, or when called out during a period of standby, and that they remain so whenever they are at work or when driving vehicles on project business.

Workers must not drink alcohol during working time. No alcohol or illegal drugs shall be consumed by workers at any of the ALIGN project work locations.

The use, possession, storage, transportation, promotion and/or sale of illegal drugs or drug equipment is forbidden during working time, in the workplace or at the worksite.

Where workers are prescribed medication or are taking over-the-counter medicines or herbal remedies that may affect their work performance or the safety of themselves or others they must advise their Employer or the organisation for whom they are providing the contracted service in addition to informing ALIGN Management.

Each Employer shall ensure that personnel under his control submit themselves for unannounced, post incident or 'with cause' testing for alcohol and/or illegal drugs if requested by ALIGN's site management. Testing shall be carried out in accordance with ALIGN "Drugs and Alcohol Policy"

Each Employer shall ensure that entry to the Site is refused to any of its visitors or employees who appear unfit through substance abuse. Illegal drugs or alcohol shall not be brought onto the site.

Anyone found in possession of illegal drugs will be reported to the police, and will be removed from site. They will be prohibited from working on any elements of the ALIGN project.

11.9 Toxic Plants

Refer to Pre-construction Information and Environmental Management Plan for the locations of toxic plants along the route and include them in relevant Risk Assessments and Method Statements.

11.10 Weils Disease and Lyme Disease

Contractors are to ensure that precautions are taken to ensure that all their personnel are made aware of the hazards associated with Weils Disease and Lyme disease and the course of action to be taken in the event of developing flu like symptoms having worked on the site.

11.11 Contact / Exposure to Agrochemicals and Fertilizers/ Animal Faeces

Contractors are to ensure that necessary precautions are taken to ensure that all their personnel are made aware of the hazards associated with coming into contact with or being exposed to agrochemicals, fertiliser, animal faeces and ensuring that suitable sanitary measures are in place such as alcohol hand washes.

12 Health and Safety File

12.1 Layout and Format

Details of ALIGN's requirements for Health & Safety File information will comply with the requirements laid out in the Construction (Design & Management) Regulations 2015 and in HS2's W1100 Health and Safety. These details will be added to this Construction Phase Plan in due course.

12.2 Arrangements for the Collection and Gathering of Information

These details will be added to this Construction Phase Plan in due course.

12.3 Storage of Information

These details will be added to this Construction Phase Plan in due course.

12.4 H&S File Approvers & Stakeholders

These details will be added to this Construction Phase Plan in due course.

12.5 Reference Documents

#	Document Title	Reference
[1]	Construction Management Plan	1MC05-ALJ-CL-PLN-C001-000002
[2]	Supply Chain Health and Safety Standard	HS2-HS2-HS-STD-000-000006
[3]	Local Traffic Management Plan	1MC05-ALJ-EV-PLN-C001-000003
[4]	Working Area Security Plan	1MC05-ALJ-HS-PLN-C001-000007
[5]	Security Strategy Plan	1MC05-ALJ-HS-PLN-C001-000001
[6]	Land Access Plans	1MC05-ALJ-GT-DSK-CS02-809507
[7]	HS2 Incident Management Procedure	HS2-HS2-CL-PLN-000-000001
[8]	Ground Investigation Layout Plan (Pages 14)	1MC05-ALJ-GT-DSK-CS02-809500
[9]	Ground Investigation-Land Access	1MC05-ALJ-GT-DSK-CS02-809507
[10]	Incident Management Plan	1MC05-ALJ-HS-PLN-C001-000041
[11]	Community Liaison Plan	1MC05-ALJ-SE-PLN-C001-000004
[12]	Community Safety Plan	1MC05-ALJ-HS-PLN-C001-000008
[13]	Contract Quality Plan	1MC05-ALJ-QY-PLN-C001-000001
[14]	Early Works Environmental Management Plan	1MC05-ALJ-EV-PLN-C001-600001
[15]	Site Emergency Preparedness and Response Plan	1MC05-ALJ-HS-PLN-C001-000032

13 Appendices

13.1 Appendix 1 - Health and Safety Policy Statement



Leadership

Using innovation and challenge to lead by example.



Safety

Putting safety and wellbeing first and creating an environment where no one gets hurt.

Integrity

Acting fairly, transparently and consistently in all of our relationships to develop trust.

Respect

Understanding others, recognising their value and behaving accordingly.



Health, Safety and Wellbeing Safety are core values of the Align JV. The Align Board of Directors is fully committed to ensuring the health, safety and wellbeing of all persons engaged in the completion of the project and others who may be affected by our activities. It is our belief and commitment that all decisions should put health, safety and wellbeing first and have a demonstrable balance in terms of legal compliance, care for people, alongside the delivery of a modern high speed railway to a high level of quality on time and within budget.



Health & Safety Vision for HS2

- We aim for the highest standards of health, safety and wellbeing so that everyone will go home unharmed and without detriment to their health and wellbeing as a result of working with us.
- That the design and planning decisions taken will enable the construction and operation of the high speed railway in one of the safest and healthiest environments of its kind.



Health and Safety 'Behaviours'



Health and Safety will be integrated into everything we do - We will develop a positive and forward thinking health, safety and wellbeing culture. It is a business requirement that we plan, create and provide safe working environments, for safe people operating safe processes established to keep everyone safe.



The management principles below underpin our core values of Health and Safety:



Respect for people - all work-related injuries, incidents and ill-health are preventable. No one is to blame when things go wrong. The wellbeing of our workforce and the proper consideration of our neighbours will help produce the legacy for which we strive.



Safe and healthy decision-making - we will raise the profile of health and wellbeing, giving each the same focus as safety. The safety, health and wellbeing of the workforce and our stakeholders will be the first consideration in all decisions that we make. We will involve appropriate experts in each construction phase so that we can demonstrate that the right choices have been made.



Making it personal - everyone involved in the project, will understand the responsibilities to themselves and others and will be actively engaged and be able to and expected to speak up if something appears to be unsafe and to intervene directly as necessary.



Align will promote the reporting and investigation of all incidents including; injuries, unsafe acts, unsafe conditions, unsafe behaviours and near misses. As a learning organisation, we are keen to learn through an open and collaborative approach to reporting and investigation; sharing the knowledge gained from incidents in order to prevent a recurrence across the project partners and the wider industry.



Align management are **responsible and accountable** for the care of their workplace, activities, people and stakeholders. We will provide our management with the required support processes and resources to achieve this.



Leadership

Using innovation and challenge to lead by example.



Safety

Putting safety and wellbeing first and creating an environment where no one gets hurt.

Integrity

Acting fairly, transparently and consistently in all of our relationships to develop trust.

Respect

Understanding others, recognising their value and behaving accordingly.



Internal health and safety specialists will actively **advise and support management** by identifying and recognising examples of good practice and areas for improvement. Performance indicators will lead in a way that enhances the performance of the HS2 project

Top-down 'Leadership'



The Align JV Board of Directors is responsible for establishing the project's Health, Safety and Wellbeing Policies and Strategy. The Project Director is responsible to the JV Board for leading the project and ensuring everyone is fully committed to those Policies and Strategies and the principles detailed within them.



Health & Safety is a Shared Commitment

All parties and personnel engaged within the Align JV and our supply chain is accountable and encouraged to visibly lead and commit to high standards of health, safety and wellbeing with strong and visible **leadership** developing trust and influencing positive **behaviours** being the foundation to this.



Continuous Improvement



Align JV's Board of Directors will actively and continually monitor the development and implementation of the Policy and procedures. We will employ the Bouygues "Indice de Vigilance" to take a **risk-based approach to assurance** of our activities, ensuring interventions are focused and targeted where improvement will have the greatest effect, an example being the focus on our twelve Basic Safety Values from the design stage through to completion of our projects



How



The above requirements will be fully integrated into all areas of the Align JV scope and expectations. Everyone is to **actively cooperate, communicate, contribute and achieve** the standards set out within this Policy Statement and our supporting plans.



Approved by:

Daniel Altier

Align Project Director

Approved by:

Chris Evans

Director Responsible for H&S

13.2 Appendix 2 - Accident and Incident Data Collection

ALIGN will collect the following data:

Fatality
Specified (Major)
>7 (3) Days
RIDDOR Injuries to non-workers (Public)
Total RIDDOR Injuries
Minor Injuries (No Lost-time)
Minor Injuries (Lost-time)
Minor Injuries (All)
Lost time (Days)
All Injuries
Hours Worked
Dangerous Occurrences
Incidents (excluding Service Strikes)
Service Strikes
Near Misses / Close Calls

ALIGN will produce a monthly, annual and rolling 12-month report on the following statistics:

Accident Frequency Rate (AFR)
12 Month Rolling AFR
Accident Incidence Rate (AIR)
Minor Injury Frequency Rate (IFR)
All Reported Injury Rate (ARI)
All Lost Time Injuries Accident Frequency Rate (LTI AFR) ¹
12 Month Rolling Service Strike Rate (SSR)
Near Miss Frequency (NMF)

13.3 Appendix 3 - Safety critical worker roles

No person (including working visitors) is permitted to carry out a Safety Critical Worker Role unless they have passed a Safety Critical Medical carried out by their employer and can provide evidence of an appropriate existing safety critical medical valid for at least six months. The minimum standard of Safety Critical Medical include:

- A baseline health questionnaire to establish any current or previous medical/psychological health history.
- Blood pressure measurement
- Height, weight and body mass index
- Audiometry (hearing test)
- Spirometry (lung function test)
- Vision
- Musculoskeletal assessment

Safety Critical Medicals are valid as follows:

Workers Age	Safety Critical Medical valid for
Up to the age of 60 years	36 months from the date of medical
Aged between 60 and 63 years	24 months from the date of medical
Aged above 63 years old	12 months from the date of medical
The above time periods between Safety Critical Medicals, are subject to no changes within the workers current state of health. Any changes in the workers state of health is to be notified to the Occupational Health Lead.	

Safety Critical Worker Roles include (SCM):		
<u>Job Title</u>	<u>What does the medical include?</u>	<u>Exceptions</u>
Tunnel Manager	PPQ/D&A/ <u>SCM</u> /SKIN/LFT/GRIP/HAVS/AUDIO	
Shift Boss	PPQ/D&A/ <u>SCM</u> /SKIN/LFT/AUDIO	
Leading Miner	PPQ/D&A/ <u>SCM</u> /SKIN/GRIP/AUDIO	
Grout Foreman	PPQ/D&A/ <u>SCM</u> /SKIN/LFT/GRIP/HAVS/AUDIO	
Grouter	PPQ/D&A/ <u>SCM</u> /SKIN/LFT/GRIP/HAVS/AUDIO	
Erector Operator	PPQ/D&A/SCM/SKIN/LFT/AUDIO	
TBM Operator	PPQ/D&A/ <u>SCM</u> /SKIN/LFT/AUDIO	
Nozzleman	PPQ/D&A/ <u>SCM</u> /SKIN/LFT/AUDIO	
SCL Worker	PPQ/D&A/ <u>SCM</u> /SKIN/LFT/GRIP/HAVS	
Tunnel TBM Miner	PPQ/D&A/ <u>SCM</u> /SKIN/LFT/GRIP/HAVS/AUDIO	
Cross Passage Leading Miner	PPQ/D&A/ <u>SCM</u> /SKIN/LFT/GRIP/HAVS/AUDIO	
Cross Passage Miner	PPQ/D&A/ <u>SCM</u> /SKIN/LFT/GRIP/HAVS/AUDIO	
Rigger	PPQ/D&A/ <u>SCM</u> /SKIN/AUDIO/LFT	
Banksman	PPQ/D&A/ <u>SCM</u> /SKIN/AUDIO/LFT	
Tunnel Electrician Foreman	PPQ/D&A/ <u>SCM</u> /SKIN	
TBM Electrician	PPQ/D&A/ <u>SCM</u> /SKIN	
Tunnel Electrician	PPQ/D&A/ <u>SCM</u> /SKIN	
Apprentice Electrician	PPQ/D&A/ <u>SCM</u> /SKIN	
Tunnel Mech. F/man Pumps STP	PPQ/D&A/ <u>SCM</u> /SKIN	
TBM Mechanical Foreman	PPQ/D&A/ <u>SCM</u> /SKIN/GRIP	
TBM Pipefitter	PPQ/D&A/ <u>SCM</u> /SKIN/GRIP	
Mechanic General	PPQ/D&A/ <u>SCM</u> /SKIN/GRIP/LFT	
Mechanical Apprentice	PPQ/D&A/ <u>SCM</u> /SKIN/GRIP/LFT	
MSV Operator(s)	PPQ/D&A/ <u>SCM</u> /SKIN/GRIP/LFT	
Traffic Marshall	PPQ/D&A/ AUDIO	
TBM Crane Operator	PPQ/D&A/ <u>SCM</u> /SKIN/ AUDIO	
Segment Feeder Operator	PPQ/D&A/ <u>SCM</u> /SKIN/AUDIO	
Drilling Rig Operator X-Passages	PPQ/D&A/ <u>SCM</u> /SKIN/LFT/ AUDIO	Audio is vital if exposed to excessive noise
Ventilation Mechanic	PPQ/D&A/SCM/SKIN/AUDIO	
MSV Crane Operator	PPQ/D&A/ <u>SCM</u> /SKIN	
Telehandler	PPQ/D&A/ <u>SCM</u> /SKIN/LFT/GRIP/HAVS/AUDIO	

Concrete Mixer Driver Dual Direction	PPQ/D&A/ <u>SCM</u> /SKIN	
Concrete Foreman	PPQ/ <u>SCM</u> /SKIN/LFT/GRIP/HAVS/AUDIO	
Concreter	PPQ/ <u>SCM</u> /SKIN/LFT/GRIP/HAVS/AUDIO	
Foreman Carpenter	PPQ	
Carpenter	PPQ/ <u>SCM</u> /SKIN/LFT/GRIP/HAVS/AUDIO	
Concrete Paver Operator	PPQ/ <u>SCM</u> /SKIN/HAVS/AUDIO	
Chief Surveyor	PPQ	
Surveyor	PPQ	
Chief Geologist	PPQ	
Geologists	PPQ	
Project Director	PPQ	
Construction Manager	PPQ	
Nurse	PPQ	
Doctor	PPQ	
Fire Marshall	PPQ	
OBC Chief	PPQ	
OBC Operator	PPQ	
Excavator /Mobile Plant Operator	PPQ/ <u>SCM</u> /SKIN /AUDIO	
HGV drivers/HIAB operator	PPQ and valid HGV licence/medical	
Working on, near or over water	PPQ/D&A/ <u>SCM</u> /SKIN/LFT/GRIP/HAVS/AUDIO	
Others	PPQ	

PPQ – Pre-Placement Questionnaire

D&A - Drug and Alcohol

SCM - Safety Critical Medical

SKIN - Skin

LFT- Lifting and Carrying

GRIP- Grip

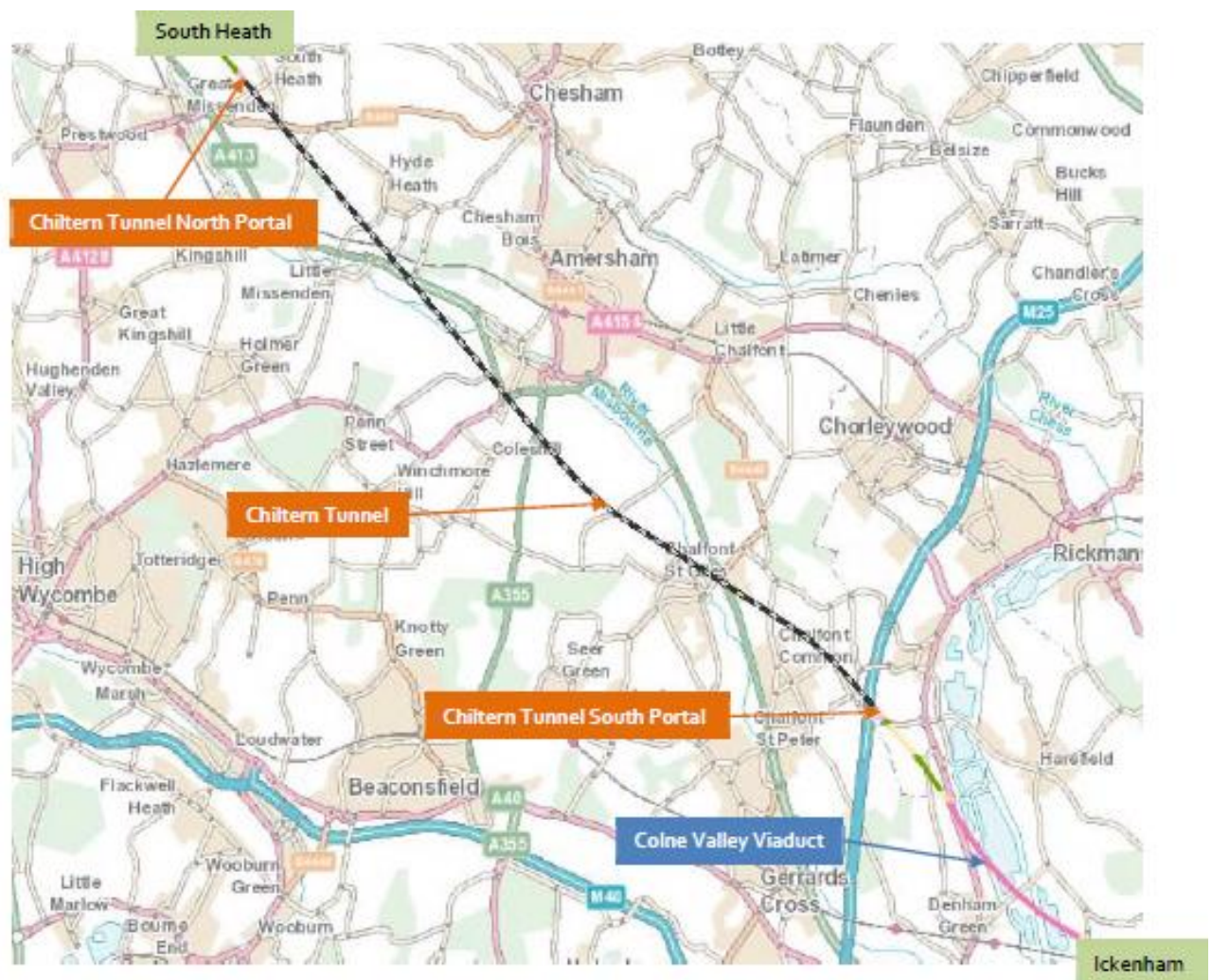
HAVS - Vibration

Audio - Noise

13.4 Appendix 4 - Site Rules & Expected Behaviours

- All personnel must have received the HS2/ALIGN site induction before they begin work. Visitors must be accompanied at all times.
- Mandatory PPE must be worn at all times. Safety footwear, hard hat, high visibility jacket/vest, trousers, gloves and light eye protection. Other items of PPE such as hearing protection/eye protection etc. should be worn on a risk assessment basis.
- Follow instructions, risk assessments and method statements.
- Do not undertake any work where there is a risk to health and safety unless you are satisfied that adequate control measures are in place.
- Permits to work are required for certain high risk activities or access to high risk areas (*permit to break ground, hot works and confined space entry*)
- Plant must only be operated by trained, competent, authorised personnel.
- Do not work at an unguarded edge without adequate fall protection.
- Always use designated pedestrian routes.
- High standards of housekeeping are required at all times.
- All accidents, incidents, near misses and spillages must be reported to your supervisor and to ALIGN.
- Horseplay or inappropriate behaviour is not acceptable on site.
- Working under the influence of alcohol or drugs is not permitted on site.
- Use of mobile phones is prohibited while driving, working at height, using machinery etc. Mobile phones must only be used only when you are in the welfare facilities or designated mobile phone location.

13.5 Appendix 5 - Project Location



13.6 Appendix 6 - CLOCS

ALIGN requires all contractors to comply with the following Undertaking and Assurance 1068

13.6.1 The assurance

U&A 1068 was given to the Cyclists' Touring Club (letter 25 November 2015) and sets out:

The Secretary of State will require the nominated undertaker to comply, as far as reasonably practicable, with the Construction Logistics for Cyclist Safety (CLOCS) standards.

The letter goes on to state that the CLOCS standard will be applied as set out within the HS2 Phase 1 Route-wide Traffic Management Plan (RTMP). The Cyclist Tourist Club (CTC) was a consultee on the development of the RTMP.

A further assurance was given the Cyclists' Touring Club (letter 14 March 2016) regarding contractors preparation of a Logistics Environment, Sustainability and Safety Management Plan (ESSMP) to demonstrate compliance and how the contractor and supply chain will seek to improve on the requirement in the RTMP.

13.6.2 Contractor requirement

The EWC works information Wl2100 states that following Royal Assent, the contractor is required to comply with the Route-wide Traffic Management Plan. As the RTMP is written in "outface facing" language, a note was prepared for issue setting out how the RTMP is to be applied (HS2-HS2-CL-ADV-000-000008). The Wl2100 is being reissued to incorporate this advice.

13.6.3 The CLOCS standard within the RTMP

The CLOCS standard sets out requirements that vehicles incorporate safety equipment to a higher standard than the legal minimum. It also covers matters such as driver training and contractor requirement to confirm that the driver is appropriately licenced.

In summary the vehicle and driver safety requirements within the CLOCS standard are:

Requirement	Comment
Work-related road risk training as a part of a drivers CPC	Generally within 3 months of regularly accessing a worksite, but can be increased as per the assurance letter
Assurance process that the driver is appropriately licenced	
All vehicles over 3.5t	
Back and side warning pictorial stickers	
All vehicles over 7.5t, additionally	

Side under run protection	Except where not capable and as per the assurance letter
Audible left turn alert	
Class IV, V and IV mirrors	
One of the following: • Side scan detection • Viewable camera system • Left turn proximity sensor • Collision avoidance system	

With regard to the fitting of the additional safety equipment, the RTMP notes that "Access to site by any UK-registered vehicle will only be permitted with the following minimum vulnerable road user safety kit fitted (to be provided by the 5th visit in any 12 month rolling period)". This is to ensure that contractors who are unaware of the requirements are not rejected on their first trips to an HS2 worksite, giving them a period to ensure compliance.

The RTMP requires that all electronic aids equipment must be working at the start of the working day and HS2 may require that all aids are working at all times.

13.6.4 Implementation

The RTMP sets out that the contractor is required to employ Duty Access Managers (DAMs) who will oversee the access to main compound and have responsibility for satellite compounds. The responsibilities of the DAMs include:

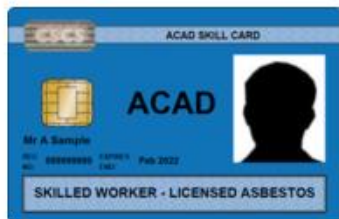
- Ensuring that vehicles and drivers comply with the project safety standards
- Rejecting vehicles, if management or safety requirements are not met (and completing "near miss records" as necessary)

The RTMP also notes that DAMs will be required to undergo necessary training for their role and responsibilities, which will be provided by the Principal Contractor.

13.6.5 Compliance team

Undertaking and Assurance 2144 requires that HS2 will employ a dedicated compliance team to assure that contractors are compliant with the project's vehicle flow and safety requirements. The assurance can be interpreted to apply from commencement Main Civil Works, but the Transport Management Team has been seeking to commence this service as soon as practicable, so that early experience can be applied to how best take this forward.

13.7 Appendix 7 - CSCS Partner Card Schemes



ACAD - Asbestos Analysts and Surveyors



ACE - Engineering Construction Workers



ADIA - Door Installers

Call 0207 788 7676



ADSA - Automatic Door Installers



ALLMI - Lorry Loaders



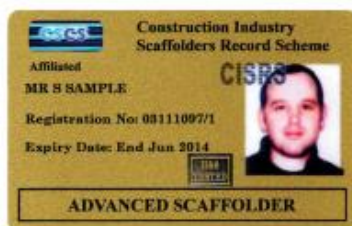
ARMI - Asbestos Removal



BFBi - Drinks Dispense Technicians



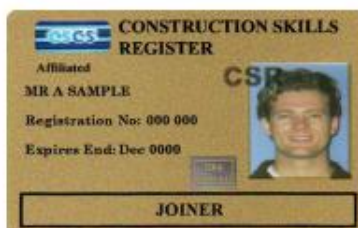
CCDO - Demolition



CISRS - Scaffolders



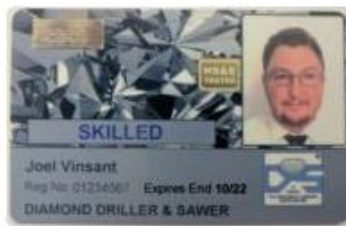
CPCS - Plant Operators



CSR - Construction and Plant Operators



CSWIP - Welding and Joining



Drilling and Sawing Association

Call 0844 879 3452



ECS - Electricians in England, Wales & Northern Ireland

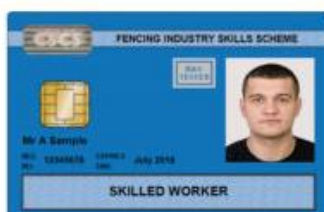


ECS - Electricians in Scotland



EUSR - Utilities

Call 0845 077 9922



FISS - Fencing Industry



GEA - Garage Equipment Installers



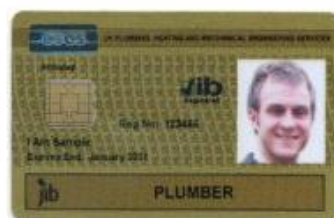
ICATS - Industrial Coating

Call 01604 438 222

Go to [website](#)



Institute of Explosive Engineers - Explosive Engineers



JIB PMES - Plumbers in England & Wales



LEEA - Lifting Equipment Engineers



LISS - Landscaping

Call 02476 690 333

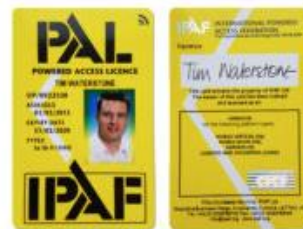


MPQC - Extractive Plant Operators



NPORS - Plant Operators

Call 01606 351240



PAL - Powered access operators



Q-card - Glazing and Construction



SAFed - Engineering Inspectors



SEIRS - Storage Equipment Installers



SICCS - Shopfitters

Call 01883 624 961



Skillcard - Heating, ventilating, air conditioning and refrigeration



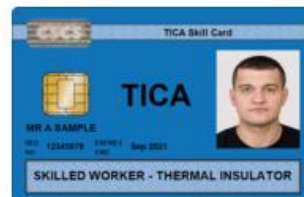
SNIJIB - Plumbers in Scotland

Call 0131 556 0600

Go to [website](#)



SSPC Trainthepainter - Coating Applicator Training & Certification



TICA - Thermal Insulation Operatives

Call 01325 466704



TTM - Temporary Traffic Management

